



FABRIC MANUFACTURING

Part: B

Course Code: FE 253-0723

*Pattern Mechanism, Chain Link, Lapping
Movement & Diagram*

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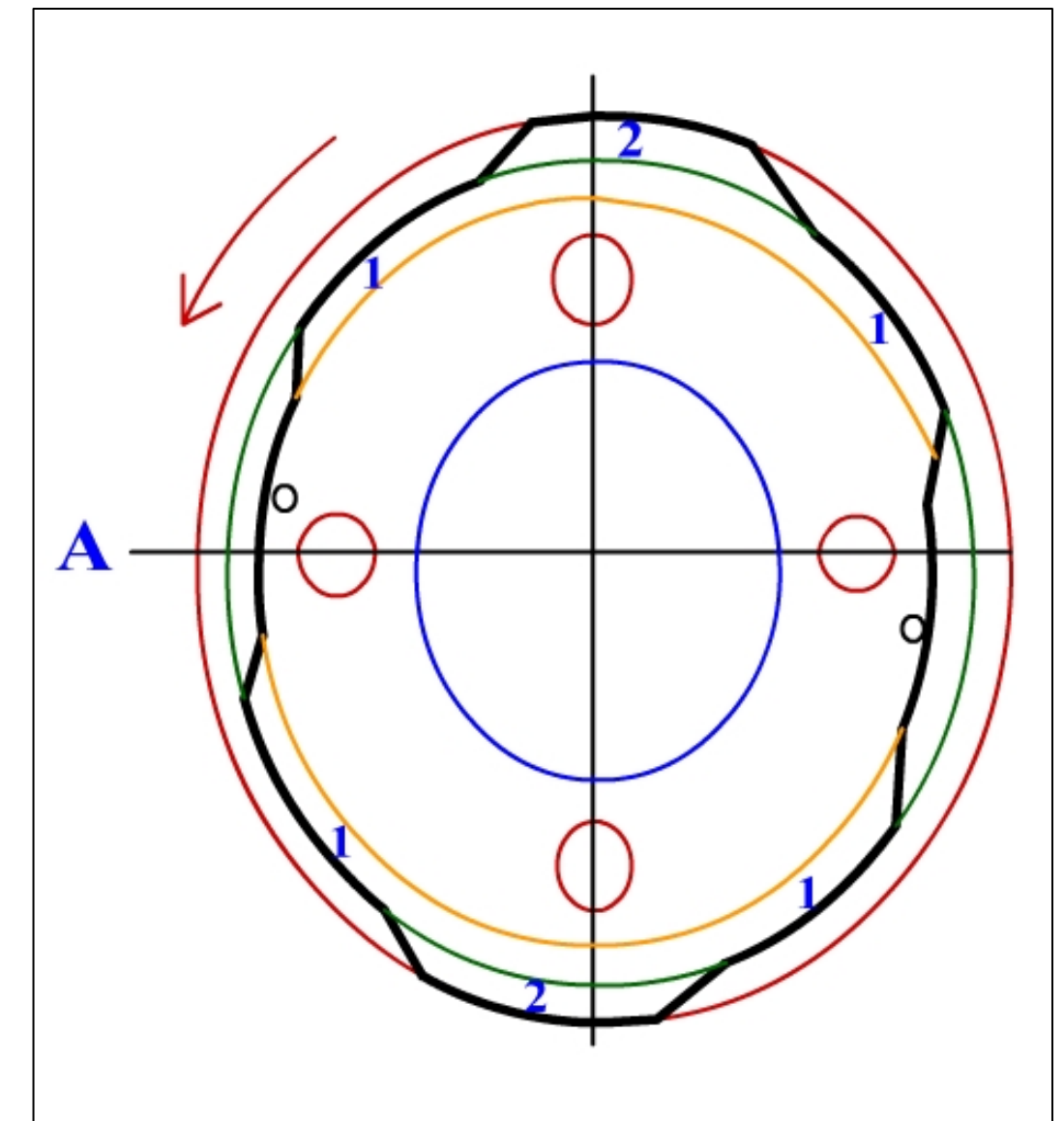
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PATTERN MECHANISM

- Swinging and shogging motions not only produce *lap of the yarns around the needles* but also *shift the yarns from one needle to other*.
- The ultimate *pattern* or structure *of the fabric* depends on the nature (direction, relative position and extent) of *movements of the guides especially by shogging motion (Overlap and Underlap)*.
- *Overlap is always done for one needle space*. If the spacing space is more than one than it may be harmful for the machine and the needle will break down.
- *Underlap basically connects two Overlap*. Underlap can be done for many spacing. Underlap is actually responsible for varying design, and it makes the fabric heavy.

PATTERN DISC

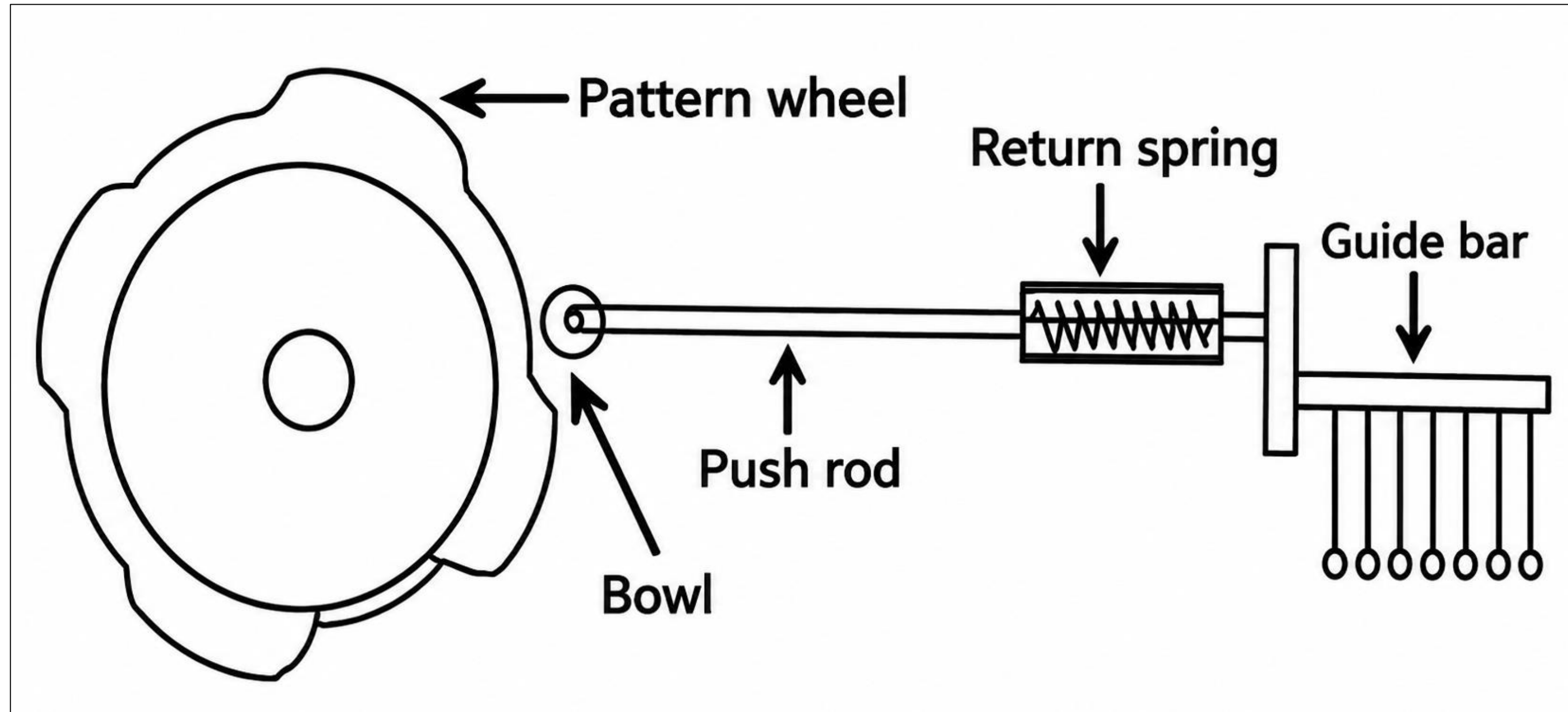
A pattern wheel is a steel disc with inclines of different slopes on its periphery. These inclines press against an anti-friction Shogging roller as the pattern wheel revolves, and the resultant displacement of the roller is transmitted to the corresponding guide bar via a push rod. In a machine there would be as many discs as the number of guide bars. All these discs are fastened together and work as a group. *As the disc can push the guide bar only in one direction, the return motion has to be generated through the action of springs.* The pattern wheel is actually a cam with inclines milled on its circumference according to the required lateral displacement profile of the guide bar.



PATTERN DISC

- A pattern disc which has different slopes on its circumference. These slopes stroke a shogging roller or bowl as the pattern disc revolves and the shogging motion is transmitted to the guide bar through a push rod.
- A pattern disc which has different slopes on its circumference. These slopes stroke a shogging roller or bowl as the pattern disc revolves and the shogging motion is transmitted to the guide bar through a push rod.
- The heights of the slopes *decide the extent of lateral displacement of the guide bars*.
- A thrust is exerted on the guide bar to shog it positively when there is a height increment of the slope of wheel.
This is called **positive shogging**.

PATTERN DISC



PATTERN DISC

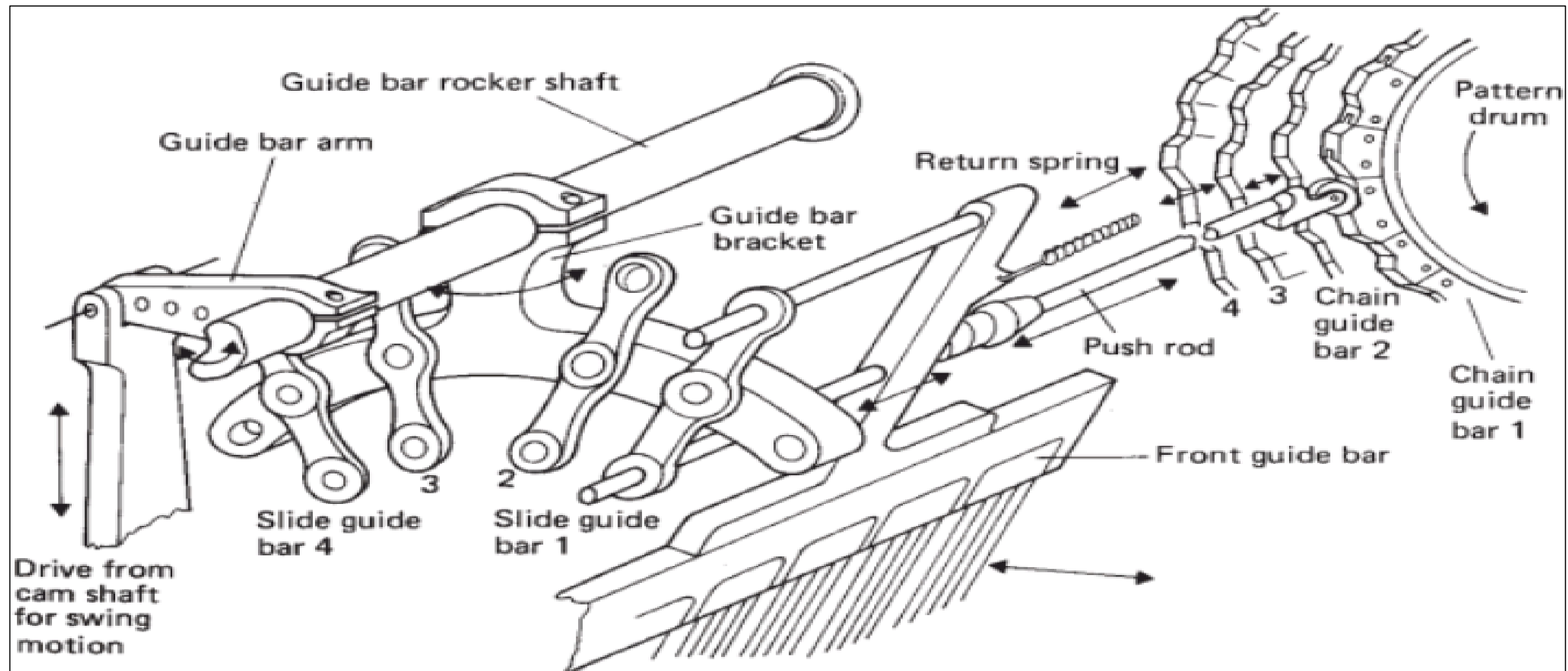


- The pattern disc is just like a cam with curves or slopes made on its circumference according to the pattern. These curves which are required for the overlapping and underlapping of the needle bar are smoothly shaped and have a well-formed transition to and from each other.

PATTERN DISC

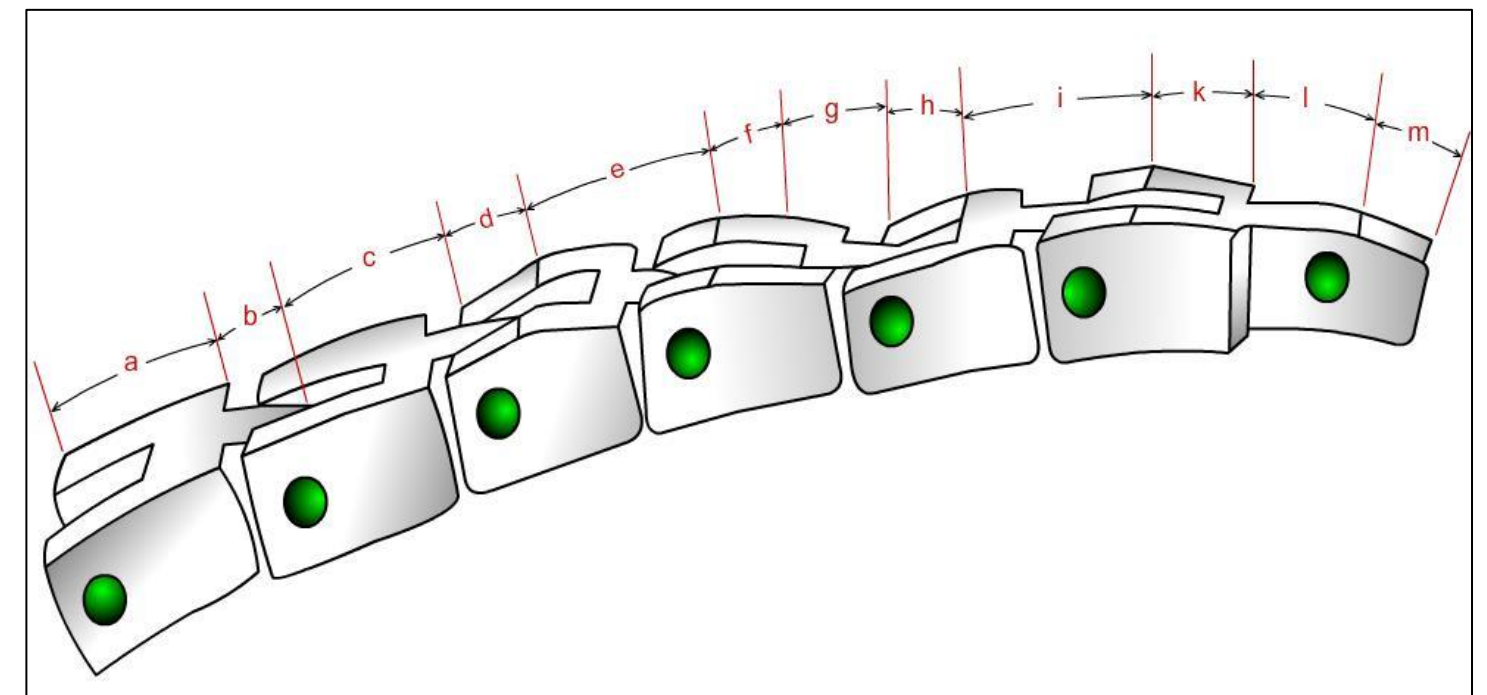
- *A decrease in height* of slope will cause the *guide bar to move towards the pattern wheel* due to the action of *return spring*. This is called *negative shogging*.
- A constant height will produce no shog and the guide bar will continue to swing through the same needle space. This is called *no shog or neutral shog*.
- Pattern wheel provides accuracy and smooth running even at high speed. A pattern wheel is economical for producing longer fabrics of simple structure. A pattern wheel has restricted utility because pattern cannot be changed to produce some other structure and it is not interchangeable between machines of different kinds.

PATTERN DRUM



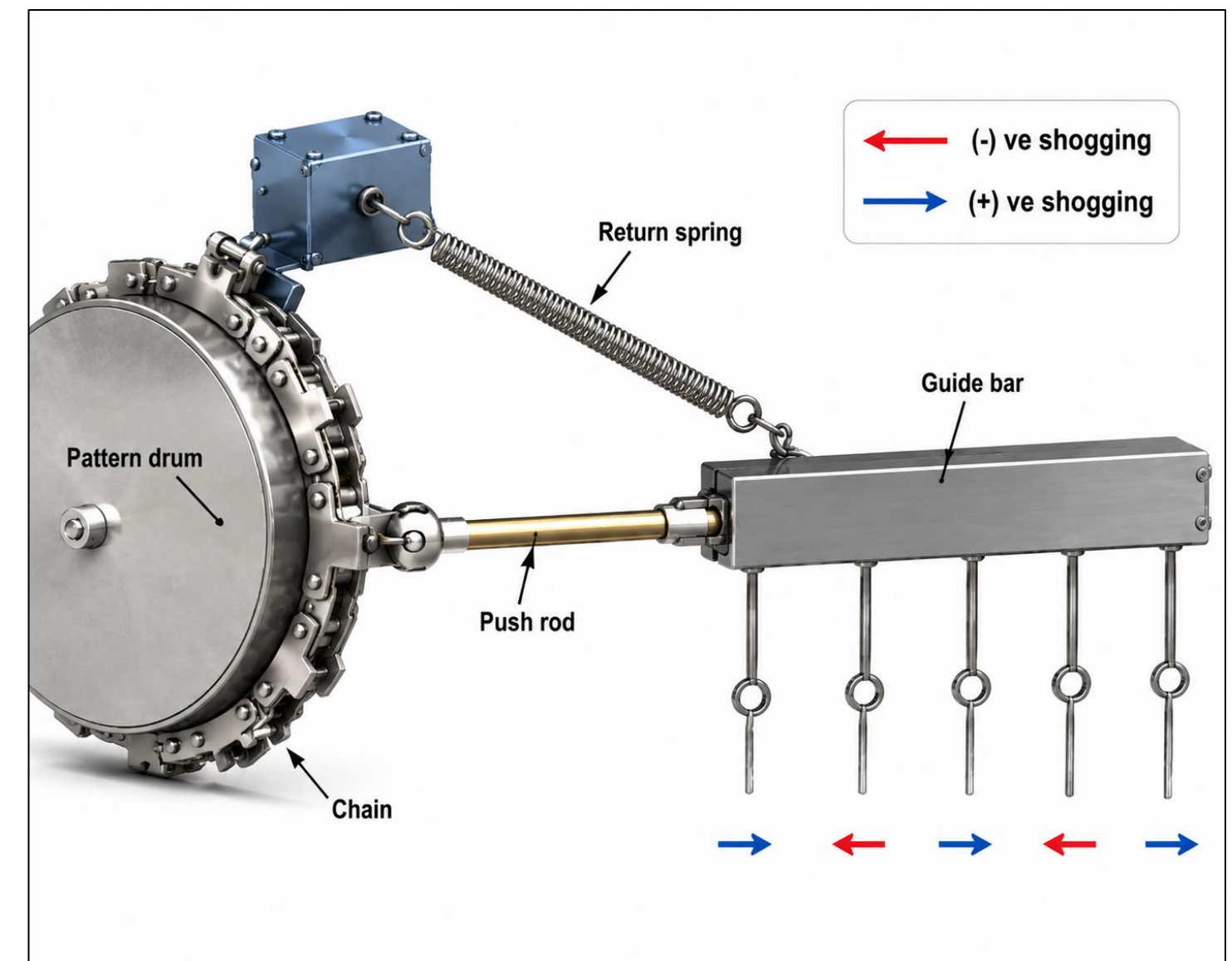
CHAIN LINK

- Chain links are *Y-shaped* parts used in the pattern mechanism of warp knitting machines.
- They look like a turning fork with a *fork part and a tail part*.
- The *tail of one link fits into the fork of the next link to connect them together*. The links are joined by pins that pass through holes in the fork and tail.
- *These pins hold all the links together and help them move as a chain*. The chain is *placed on the pattern drum*. When the drum rotates, the chain links move forward at the correct time and control the guide bar movement.



CHAIN LINK

- An increase in height one link to the next produces a thrust against the end of the guide bar, *shogging it positively* into the machine.
- A decrease will produce a negative shog towards the pattern shaft as the result of the action of a return spring.
- A constant height will produce no shog if the guide bar will continue to swing through the same needle space.
- The periphery of the pattern wheel or chain track is scanned by a roller which is linked by a flexible jointed push-rod to the end of the guide bar.
- The drive for the pattern shaft is obtained from the main cam-shaft.



CHAIN LINK

- A pattern chain is constructed by connecting a large number of individual pattern links and placed over a pattern drum. The shogging roller moves in contact with this chain and transfers required motion to the guide bar.
- *Each pattern link is equivalent to a curve in the pattern disc.*



CHAIN LINK

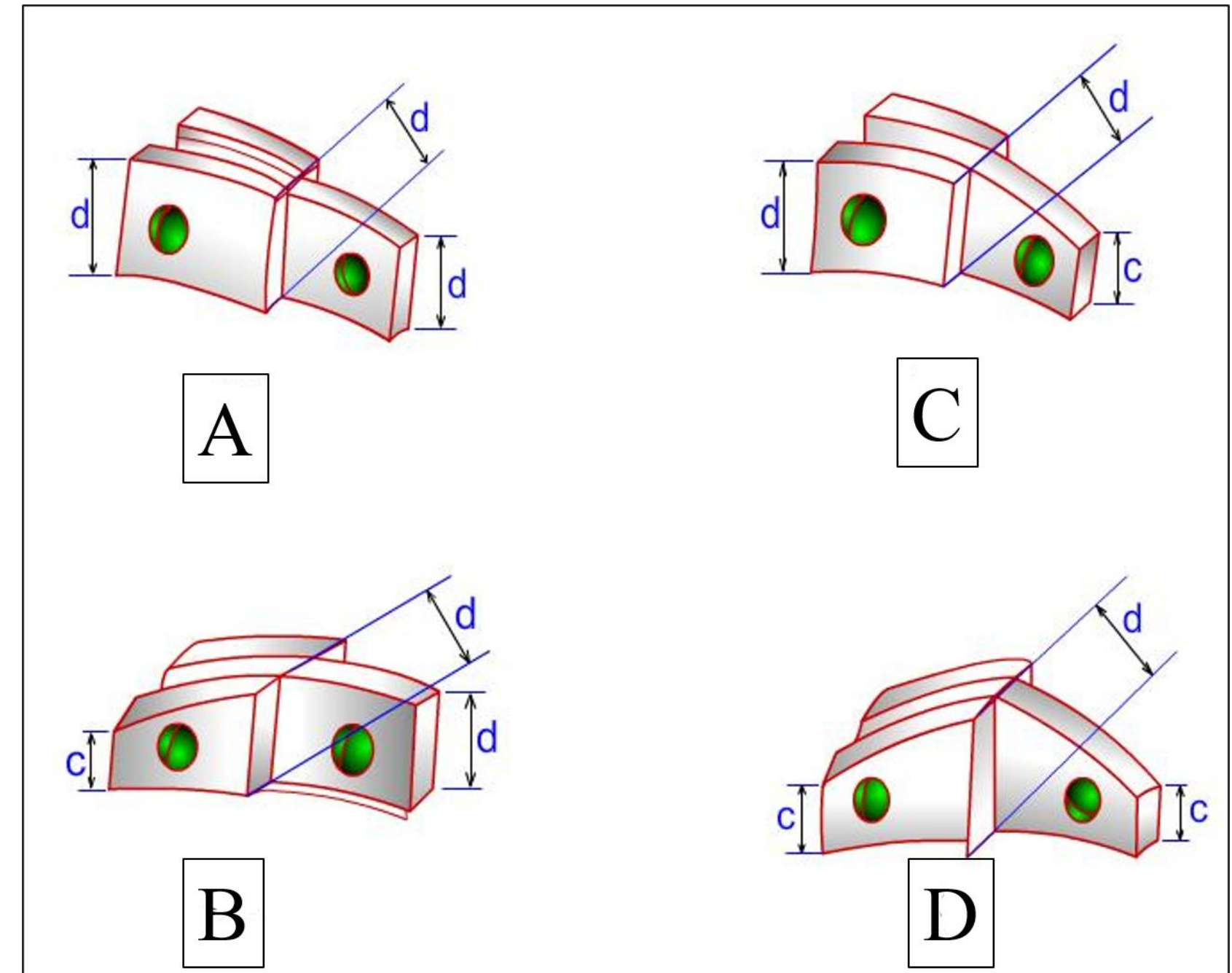
- The chain link is slightly curved so that it can fit properly on the surface of the pattern drum.
- The fork side of the link is the front part when connecting one link with another.
- *Links are made according to the machine gauge.*
- To avoid confusion, the *gauge number is marked on the links.*
- The height of the link is adjusted to create the required slope.
- The **angle** and **length of the link** edge must be very **accurate to ensure correct shogging** movement timing.
- A special grinding gauge is usually supplied by the machine manufacturer, and links are ground according to that gauge.
- Pattern links are also available with pre-ground edges for different chain arrangements.
- These links are identified by letters according to their ground edge shape.

CHAIN LINK

- “A” - Unground Link
- “B” - Fork is Ground
- “C” - Tail is Ground
- “D” - Both Fork & Tail Ground

The shogging movement is initiated by *varying the radius of the continuously turning pattern shaft* either in the form of different heights of pattern links.

It passes over a pattern drum attached to the shaft or in the form of carefully shaped solid metal circular cams, termed pattern wheels, attached to it.



CHAIN LINK

- The links have also different heights representing the amount of guide bar movement in terms of needle spaces, being numbered 0,1,2,3,4 and upwards.
- The chain is prepared by laying the required number of links side by side on a flat surface and then are joined by inserting on the holes provided for the purpose.



CHAIN LINK

- For smoother movement of the guide bar, the links need precision grinding at the points where they connect.
- As the number and the shape of the pattern links in a pattern chain can be changed, the pattern range is greater, and the technique is more versatile.
- The *chain links vary in height* by multiples of machine gauge, i.e., on *22G machine by multiples of 1/22 inches*. Each link is generally stamped with the gauge and with a number, e.g. 0, 1, 2, 3, 4 etc. *Link gauge is equal to machine gauge*.
- The lowest value link (size 0) will place the guide bar nearest to the pattern drum.

CHAIN LINK

- Links of size 1 are *1/22 inch higher* than the links of size 0, and will position the guide bar 1/22 inch, i.e. *one needle space, far*.
- Progressively, as the link number increases, *guide bar will be positioned further away from the pattern drum*. Thus, a link of height 4 will cause the guide bar to take up a position of *4/22 inch (four needle spaces)* further away from the *drum than a link of height 0*.
- Machine RPM = Course Per Minute (CPM)
- RPM of pattern drum = Machine RPM/16. So, if CPM is 1600, then 1600 courses will be produced after 16 times revolution of pattern drum.

CHAIN LINK

22 Gauze link thickness

Let, the thickness of size 0 = x''

So,

$$\text{Size 1} = x'' + (1/22)''$$

$$\text{Size 2} = x'' + (2/22)''$$

$$\text{Size 3} = x'' + (3/22)''$$

$$\text{Size 4} = x'' + (4/22)''$$

32 Gauze link thickness

Let, the thickness of size 0 = x''

So,

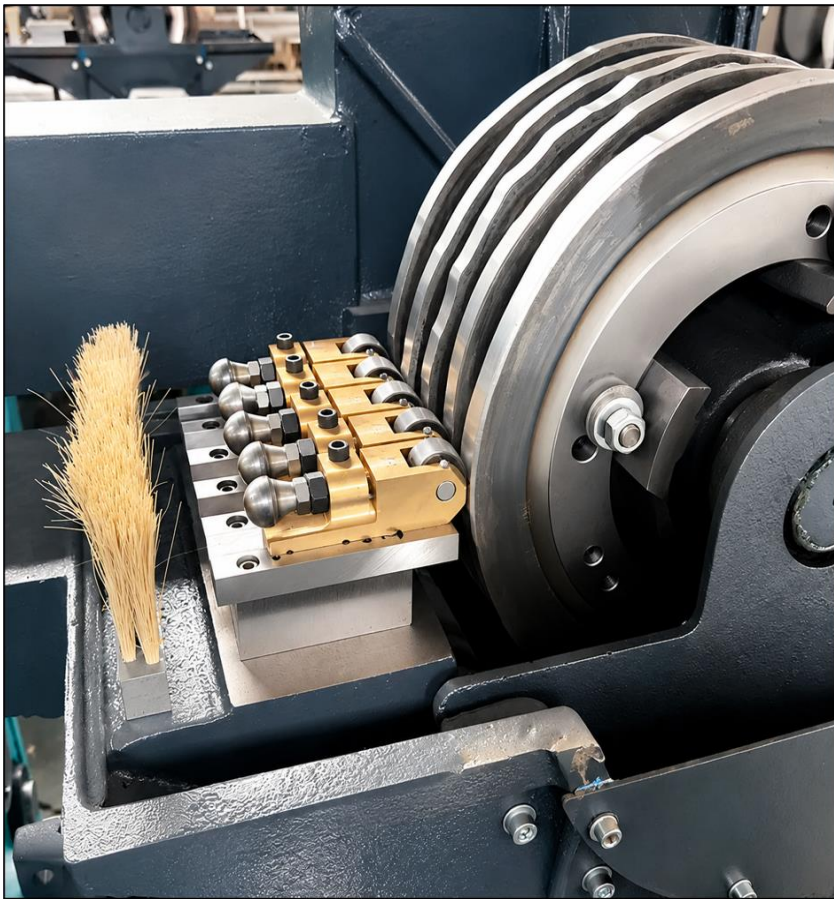
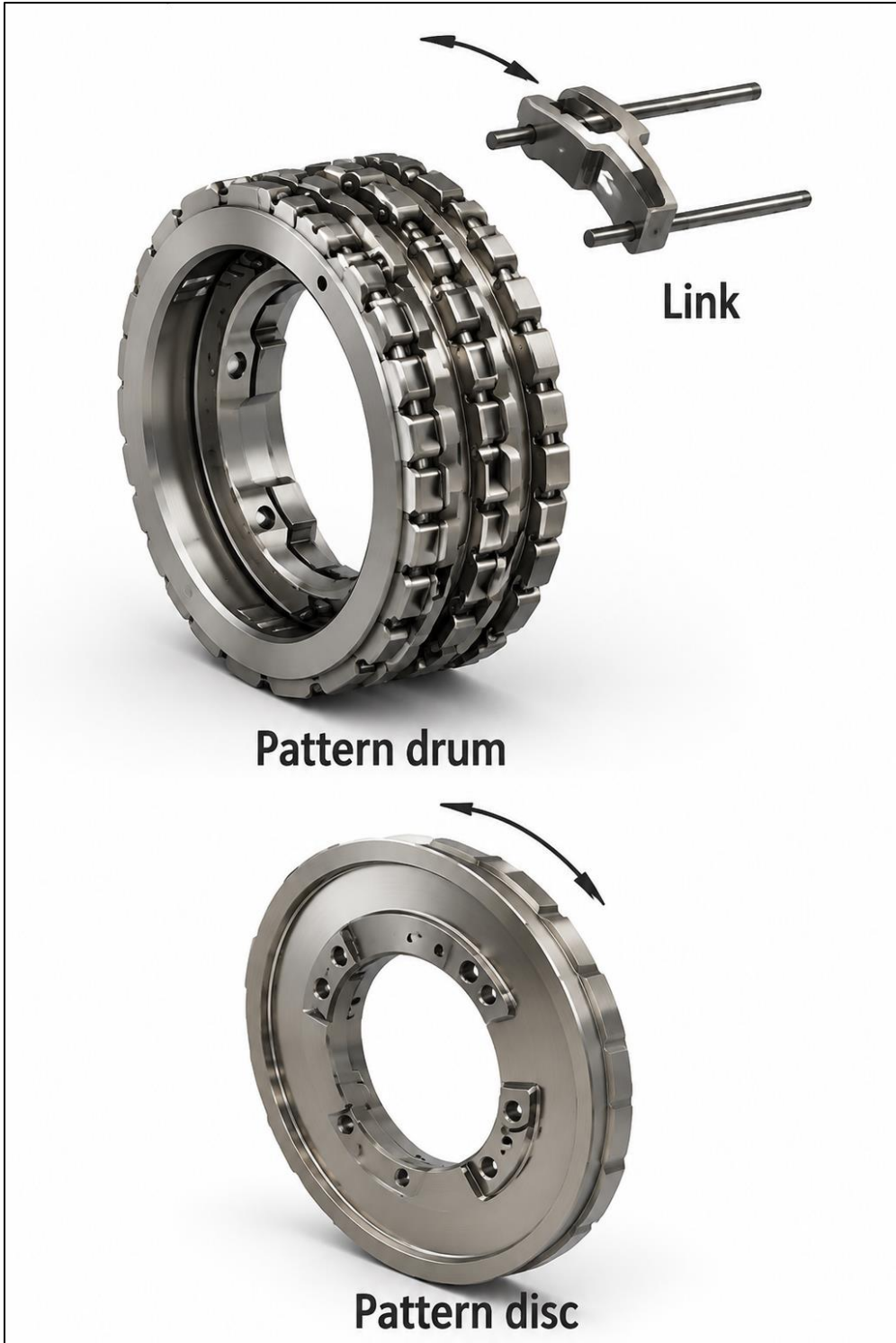
$$\text{Size 1} = x'' + (1/32)''$$

$$\text{Size 2} = x'' + (2/32)''$$

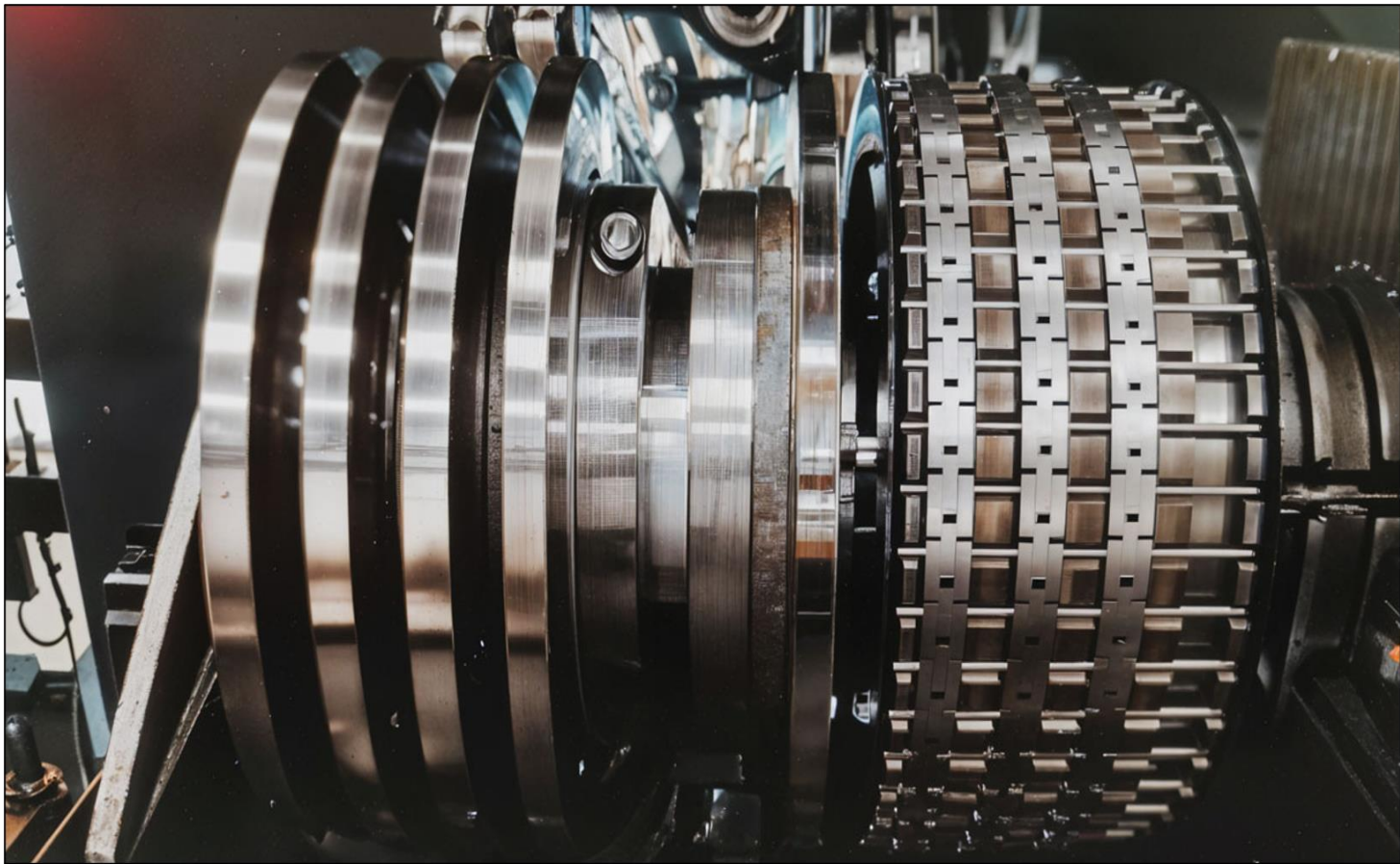
$$\text{Size 3} = x'' + (3/32)''$$

$$\text{Size 4} = x'' + (4/32)''$$

PATTERN DRUM VS PATTERN DISC



Pattern Disc

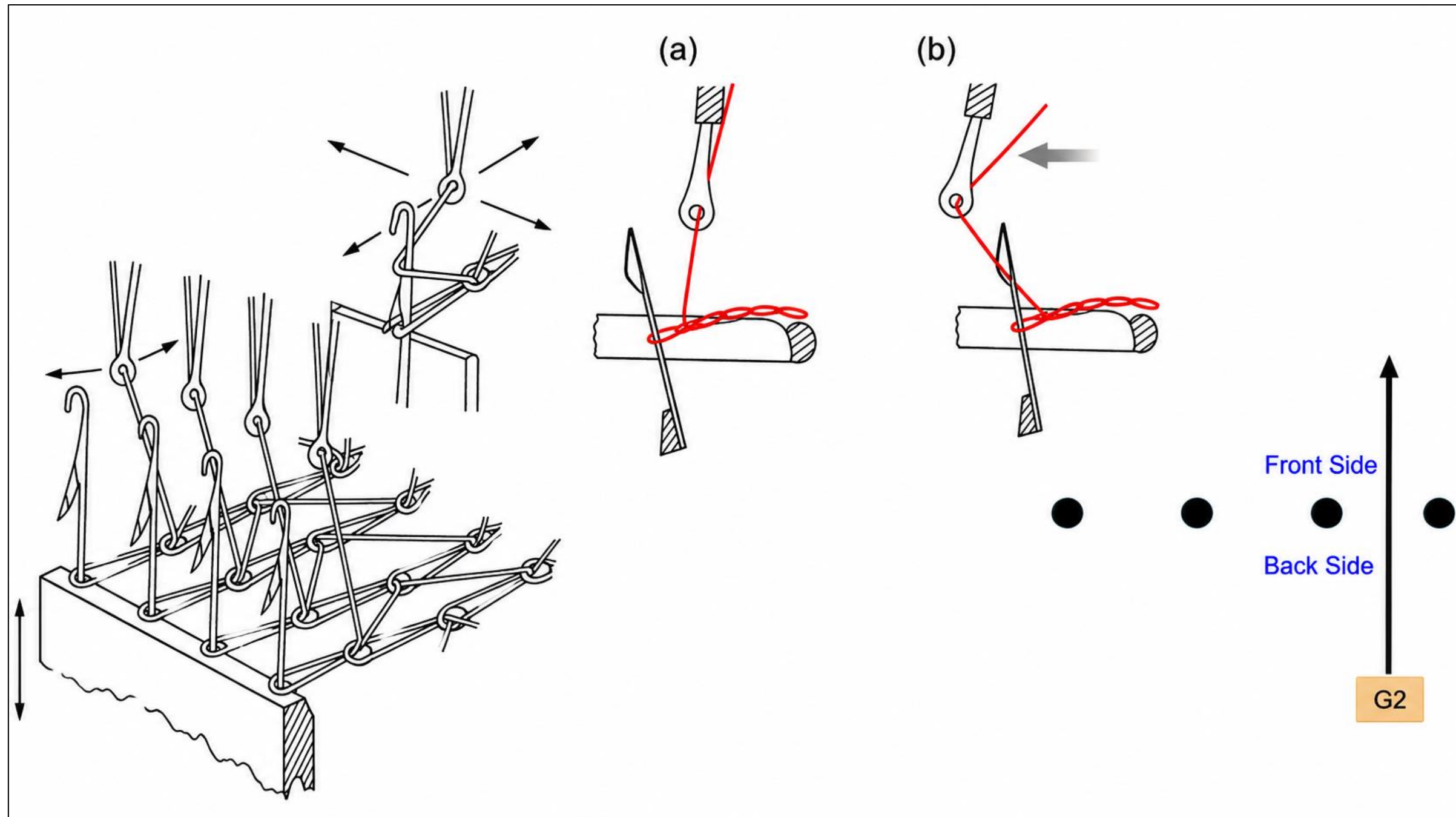


Pattern Drum with Link

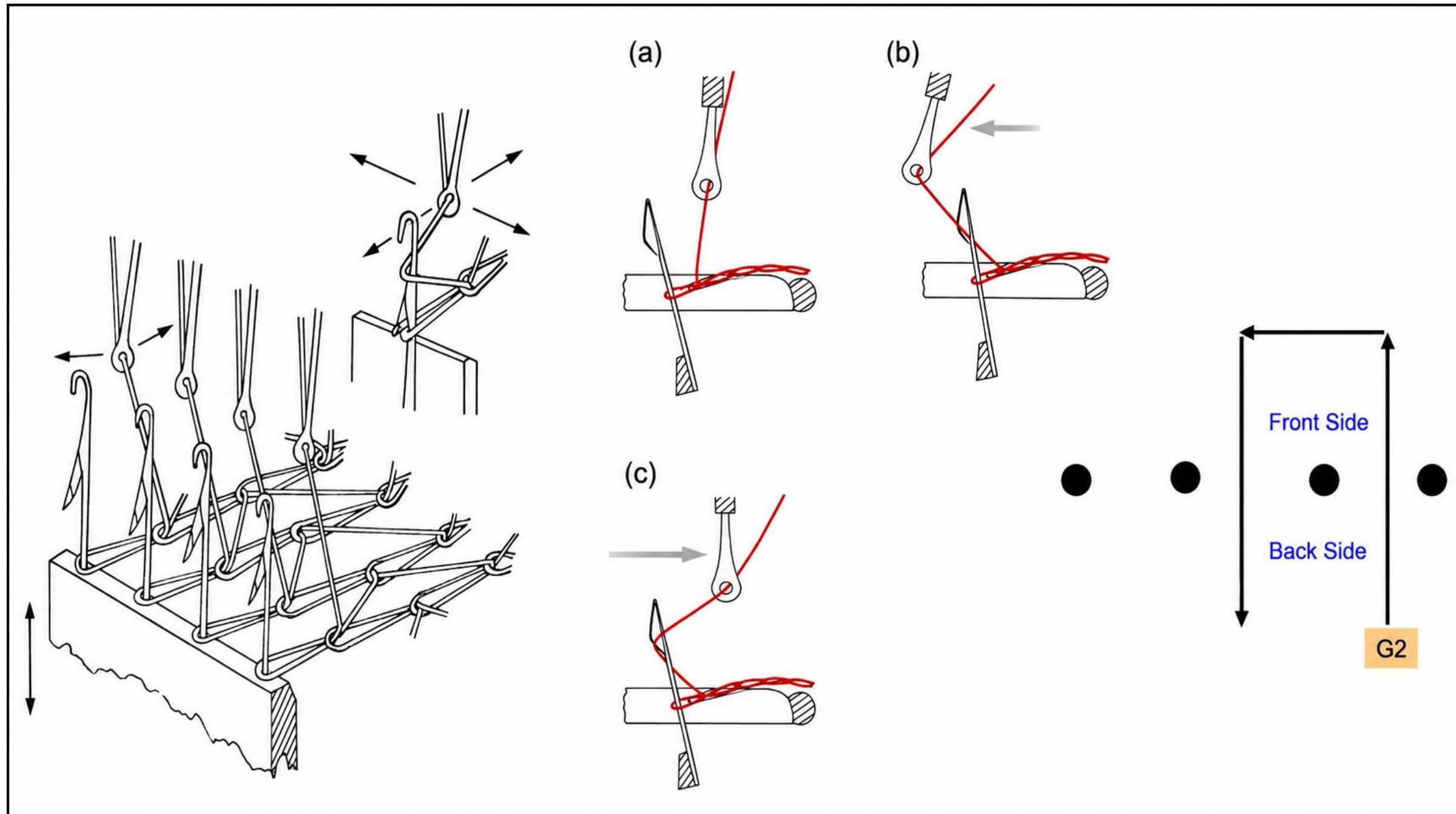
PATTERN DRUM VS PATTERN DISC

Pattern drum	Pattern disc
1. Pattern drum is normal cylindrical drum whose surface is grooved for attaching chain link	1. Pattern disc is a carefully shaped metal circular cams
2. There is an arrangement for chain link mounting	2. No such arrangements
3. Different design can produce by changing the chain link over the drum surface	3. For different design different shaped disc used
4. Design pattern is controlled by the chain link over the drum	4. Design pattern control by the engraved portion of the disc
5. Suitable for fancy structure production	5. Only for simple construction or small repeated pattern
6. Less costly	6. More cost for engraving
7. Higher jerky movement	7. High accuracy and smooth running
8. Economical for short run and long run production	8. Economical only long run production of common simple repeat structure

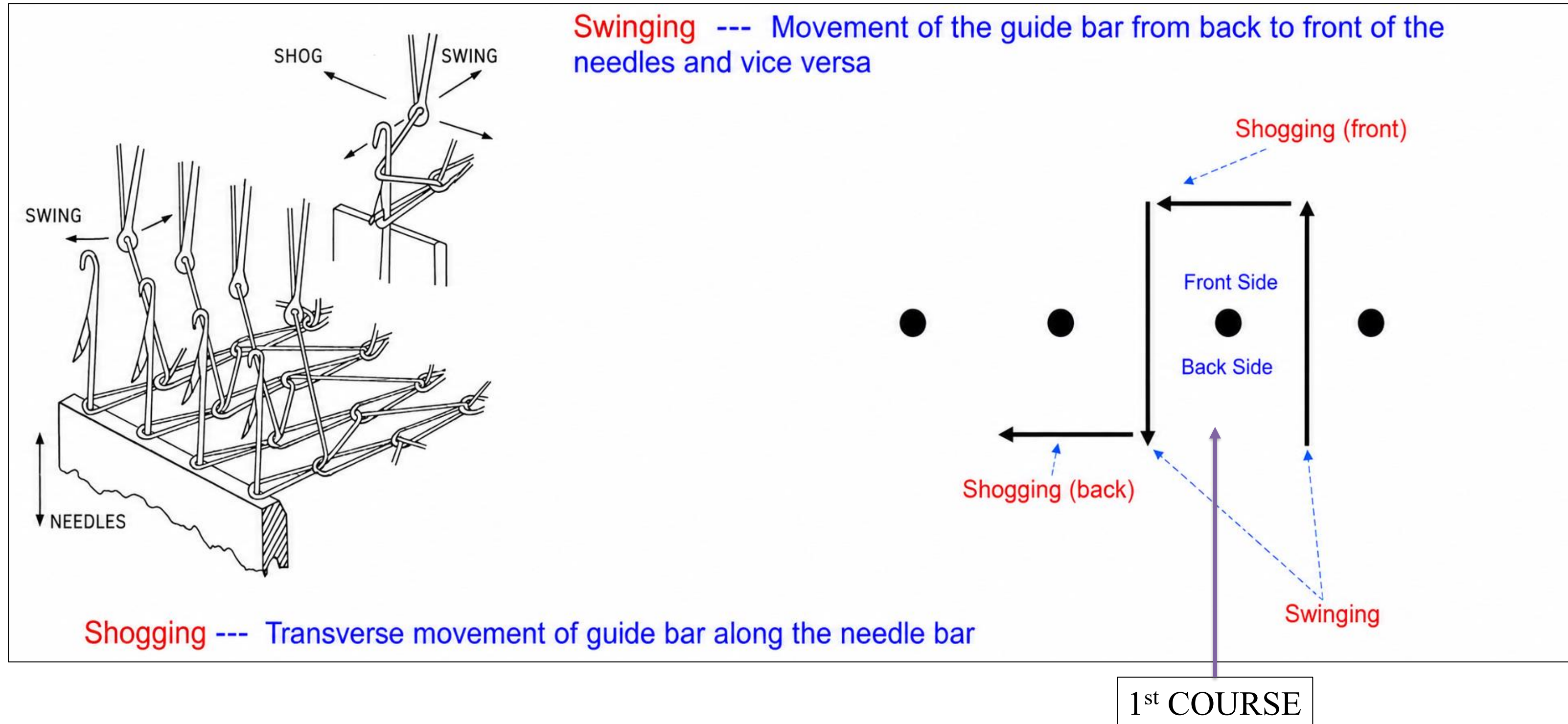
GUIDE BAR MOVEMENT



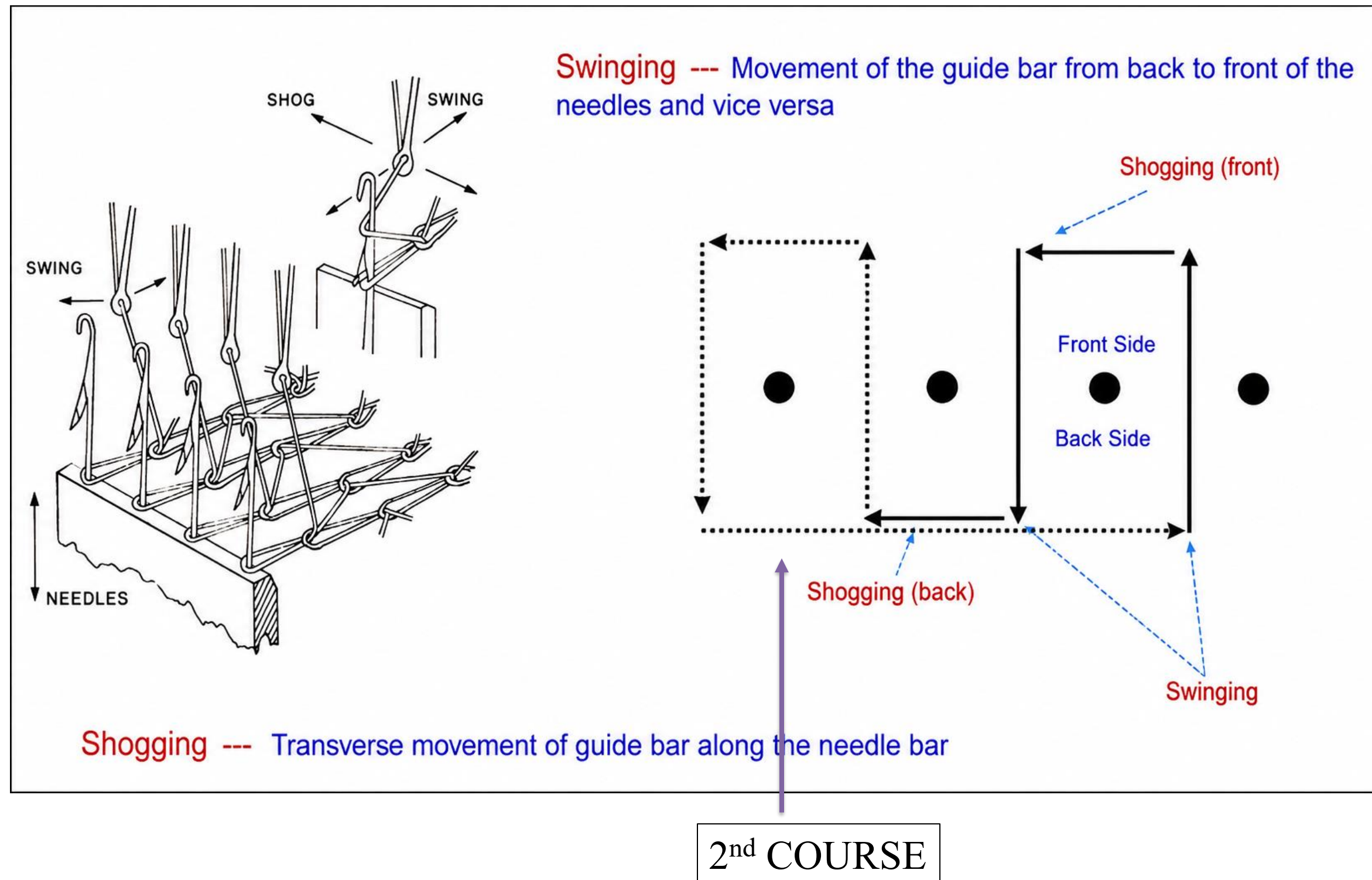
GUIDE BAR MOVEMENT



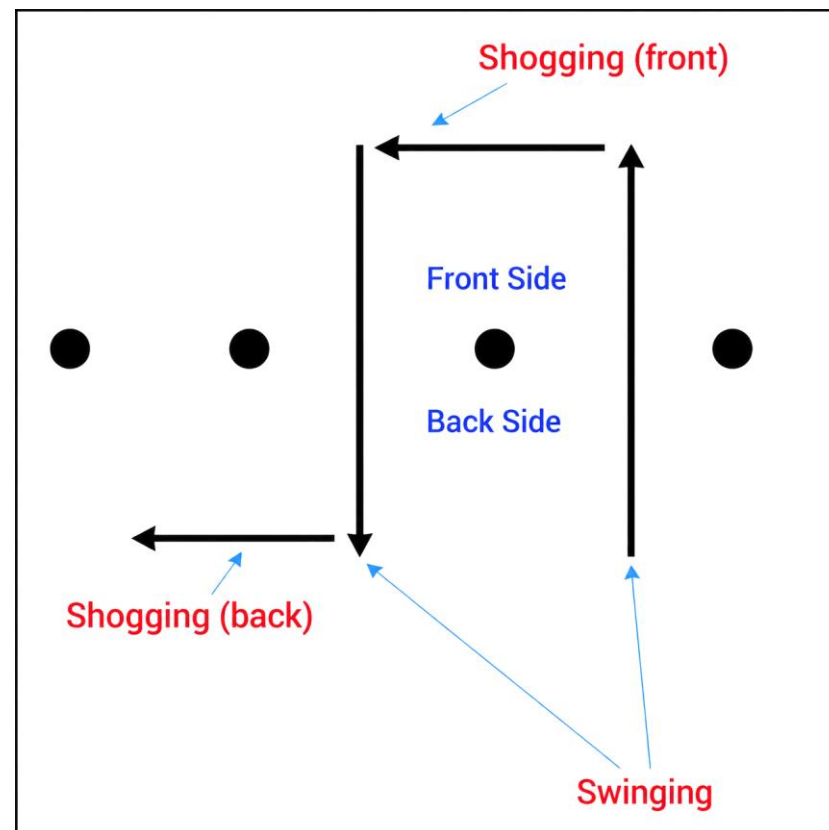
GUIDE BAR MOVEMENT



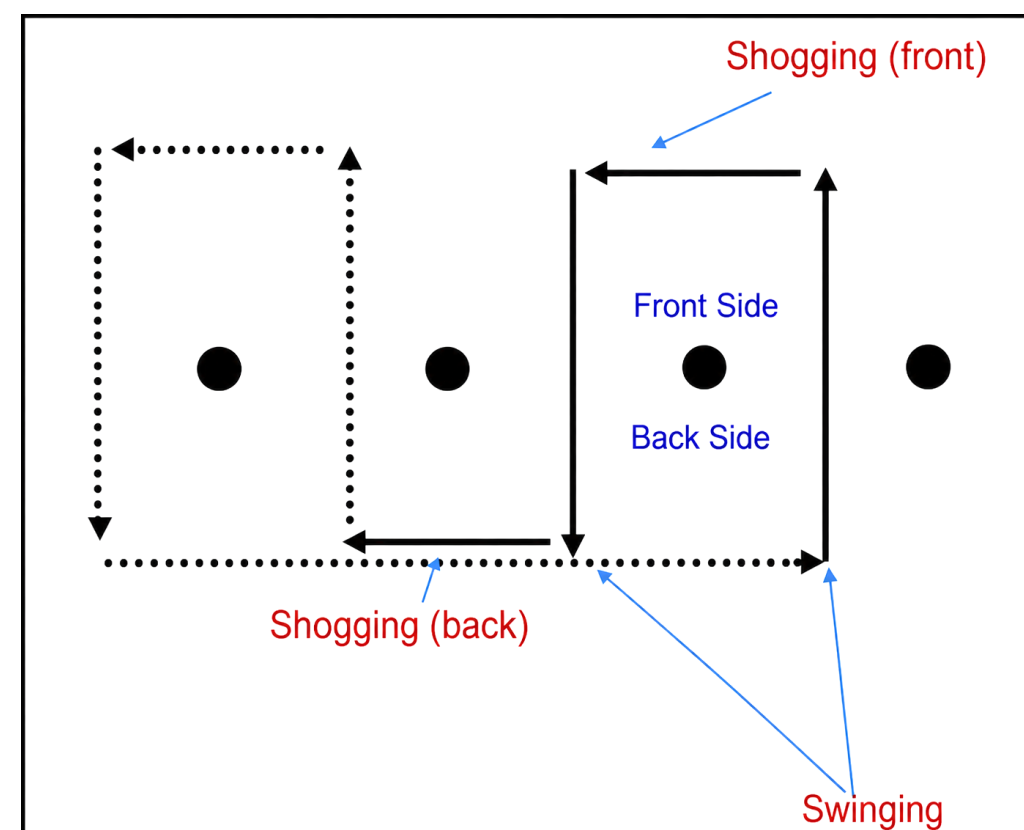
GUIDE BAR MOVEMENT



GUIDE BAR MOVEMENT



1st COURSE



2nd COURSE

- This is how a single guide bar supply yarn in two needle in two consecutive courses.
- Two needles are getting yarn by same guide.

GUIDE BAR MOVEMENT

2 Swinging

1. Back to Front
2. Front to Back

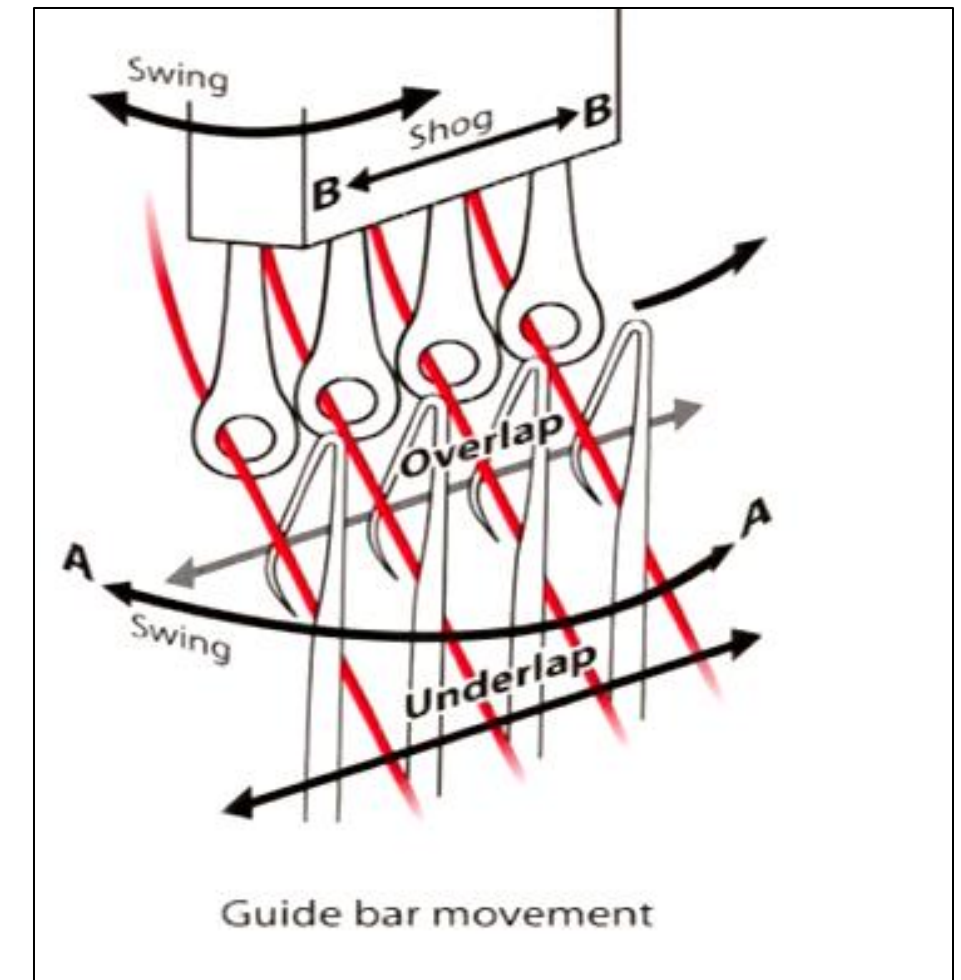
2 Shogging

1. Overlap (Front)
2. Underlap (Back)

Overlap: 1 Pitch

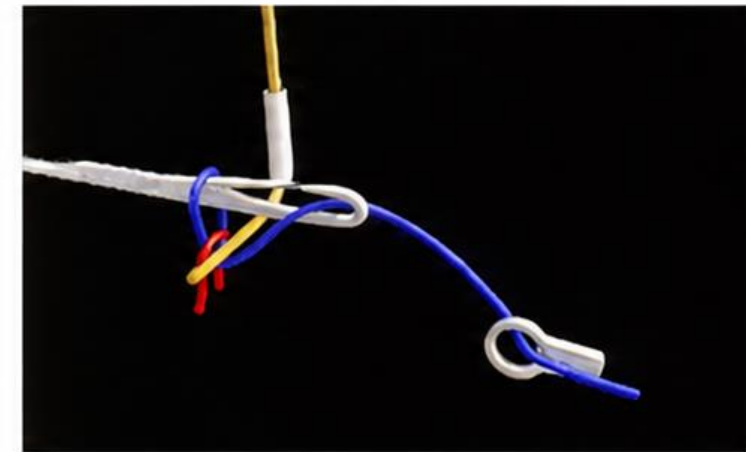
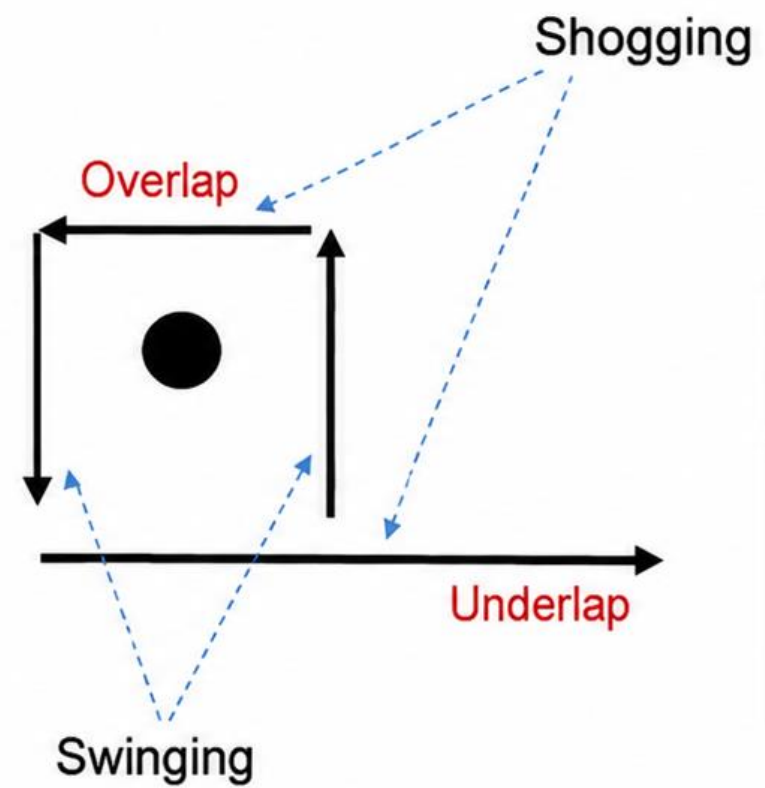
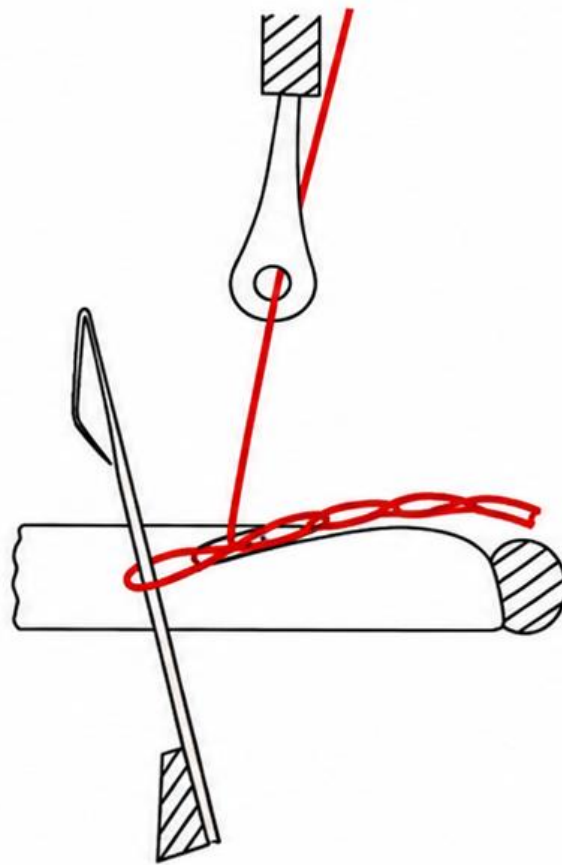
Underlap: Varying (1,2,3 Pitch etc.)

Pitch: Distance between two consecutive needles.



GUIDE BAR MOVEMENT

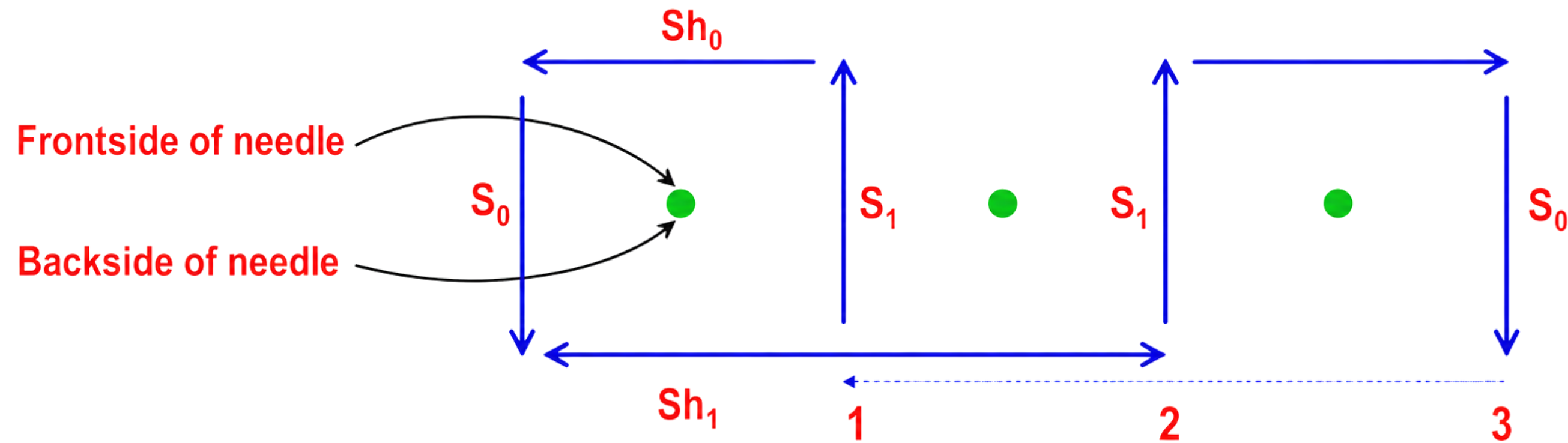
Swinging and Shogging



Overlap – 1 Pitch

Underlap – Varying (1, 2, 3 Pitch, etc)

GUIDE BAR MOVEMENT



Swinging (S_1) ---- From back to front

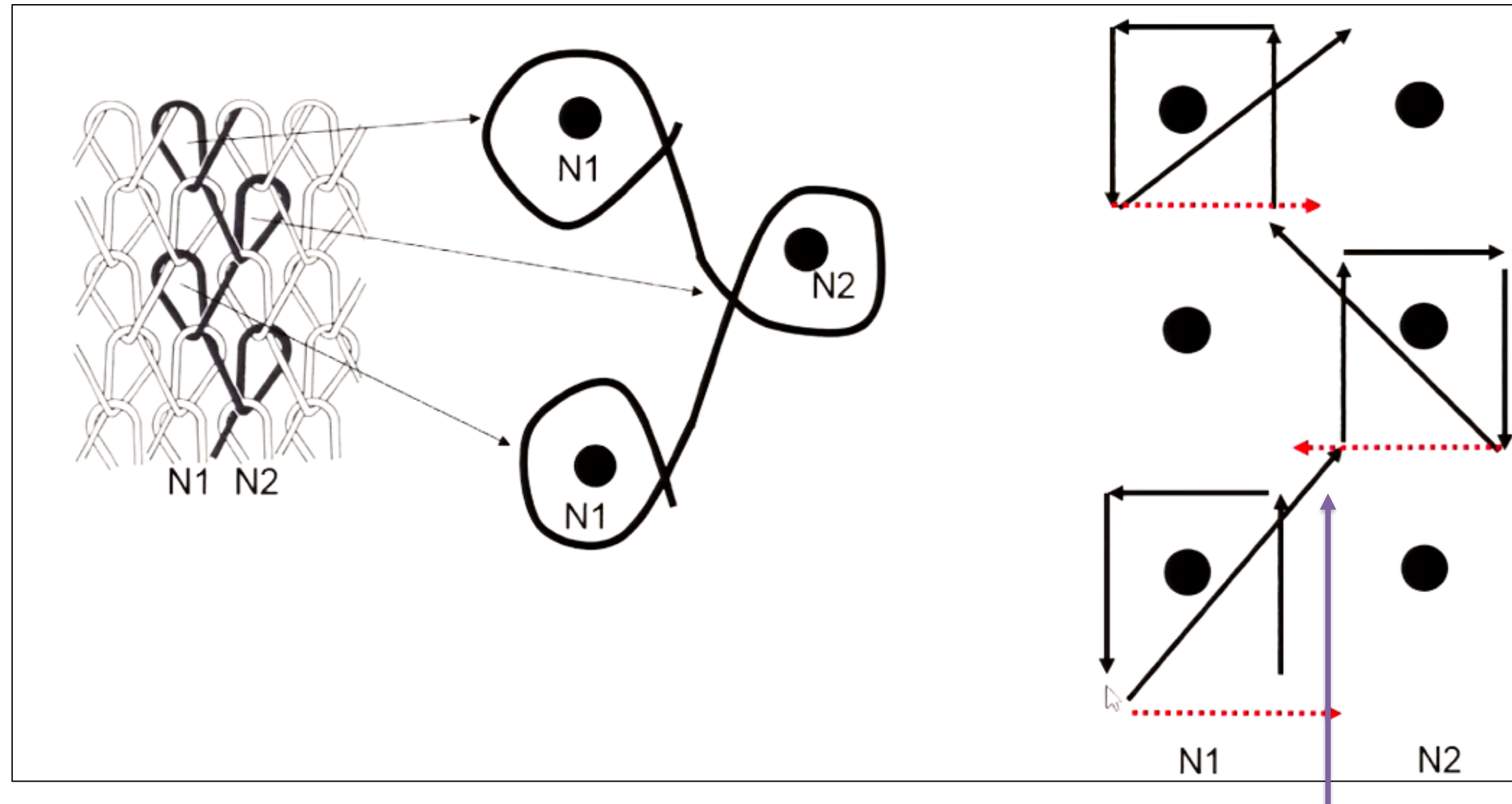
Swinging (S_0) ---- From front to back

- Overlap and Underlap are in opposite direction in same course.
- Overlap = 1 Pitch; Underlap = 2 Pitch
- Both Overlap & Underlap for two alternating courses are in opposite direction.

Shogging (Sh_0) --- in the front of the needle (also called as overlap)

Shogging (Sh_1) --- in the back of the needle (also called as underlap)

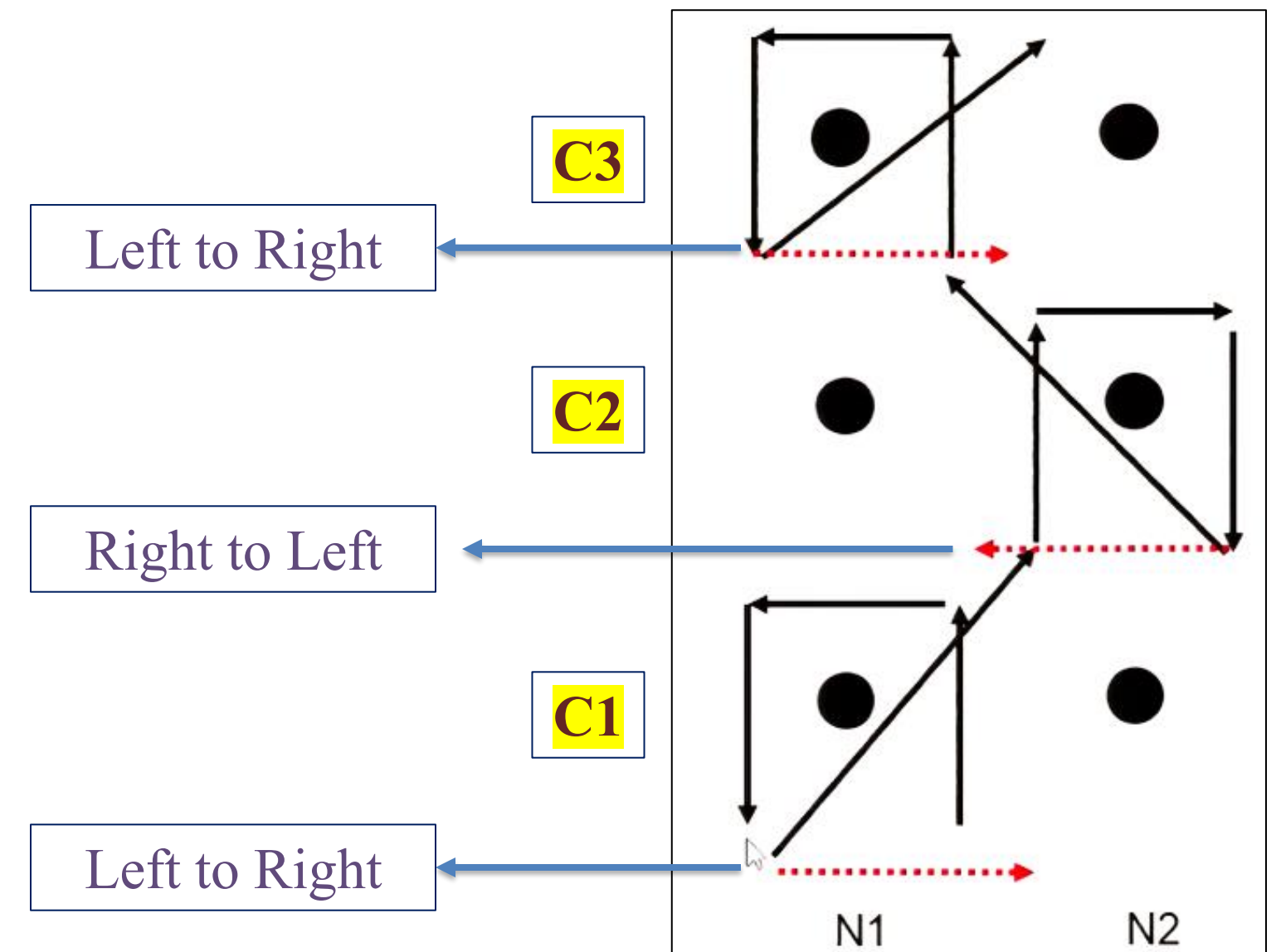
GUIDE BAR MOVEMENT



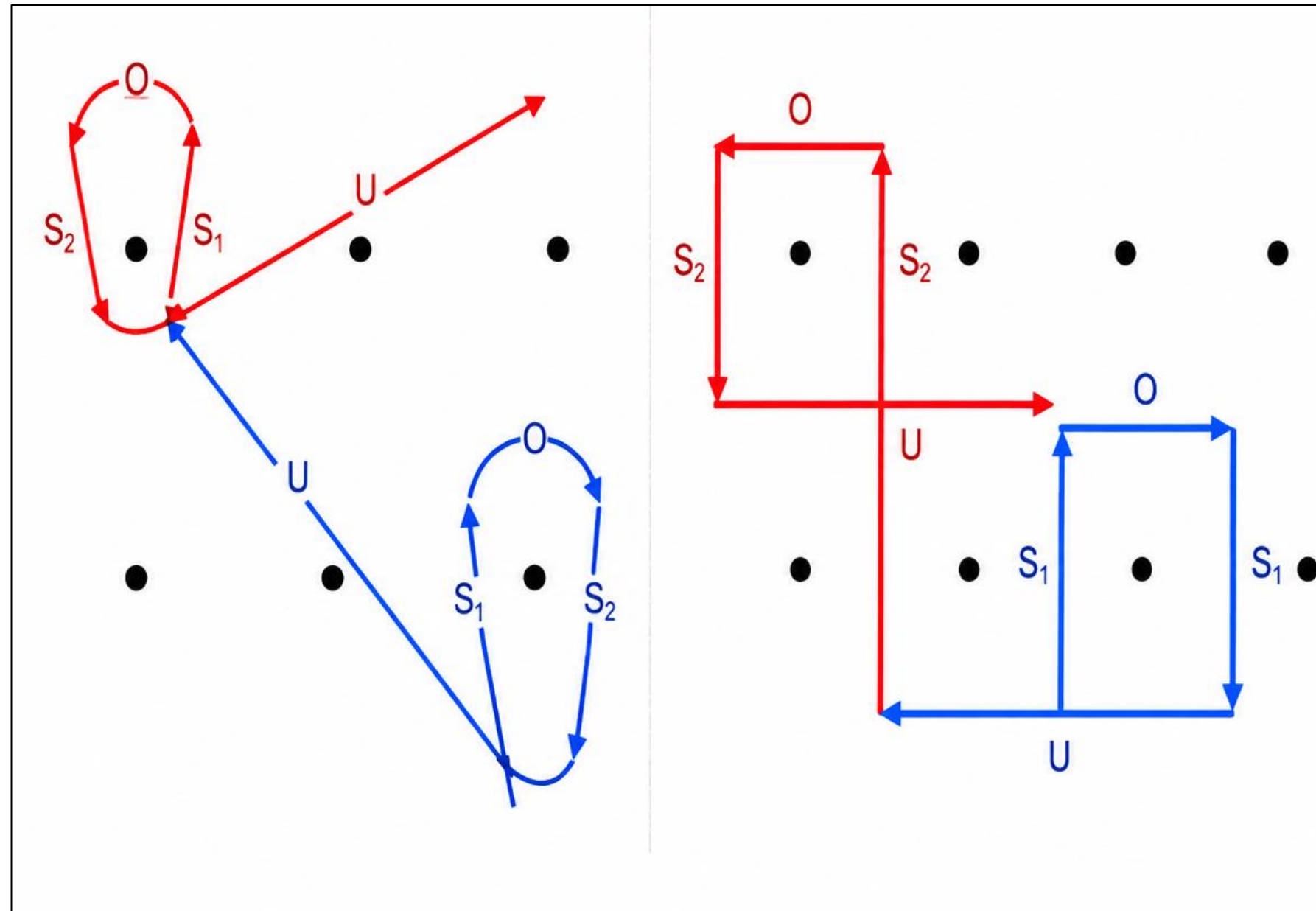
NORMALLY GUIDE BAR DOESN'T MOVE DIAGONALLY

GUIDE BAR MOVEMENT

Both Overlap & Underlap for two alternating courses are in opposite direction. C1 & C2.



GUIDE BAR MOVEMENT



Actual Fabric View

Plane View

LAPPING DIAGRAM

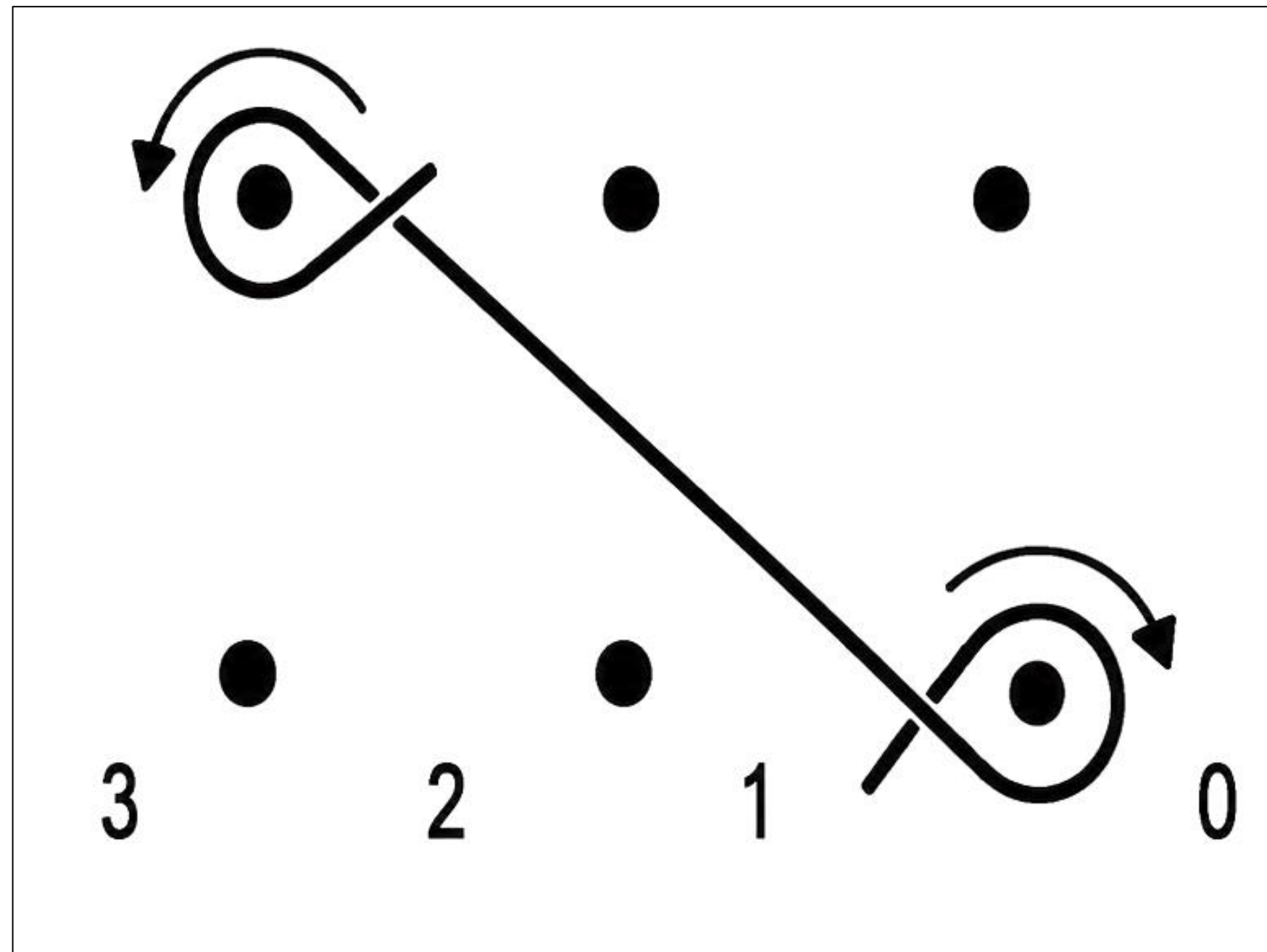
- Lapping diagram are drawn around *horizontal rows of points that represent needles in plain view*.
- Usually *assuming the pattern mechanism to be on the right*.
- As the guides position themselves in the *spaces between the needles* the positions between vertical columns of points can be given *chain link numbers commencing with the 'O' position which is to the right of the right-hand column of points*.
- Provided the direction and extent of the overlaps are correctly indicated in the lapping diagram and chain notation.

LAPPING DIAGRAM

When plotting a lapping diagram, a few basic rules must be observed:

- When the fabric is composed of *more than one guide bar, the lapping movement of each bar must be represented separately by one thread.*
- The lapping diagram of all bars must be plotted from the same course, so that the relative lateral position of all is kept.
- *The numbers allocated for the spaces between the needle must always correspond to the position of the pattern mechanism.*
- When mounting a chain on the pattern drum, special care must be taken to ensure the position of the chain corresponds to the movement of the knitting elements.

GUIDE BAR LAPPING DIAGRAM



CHAIN NOTATION

A Chain Notation is a list in correct sequence of chain link numbers spaced in to knitting cycles for each guide bar necessary to produce a particular structure.

O	U	O	U
1-0	/	2-3	/

O = Overlap and U = Underlap

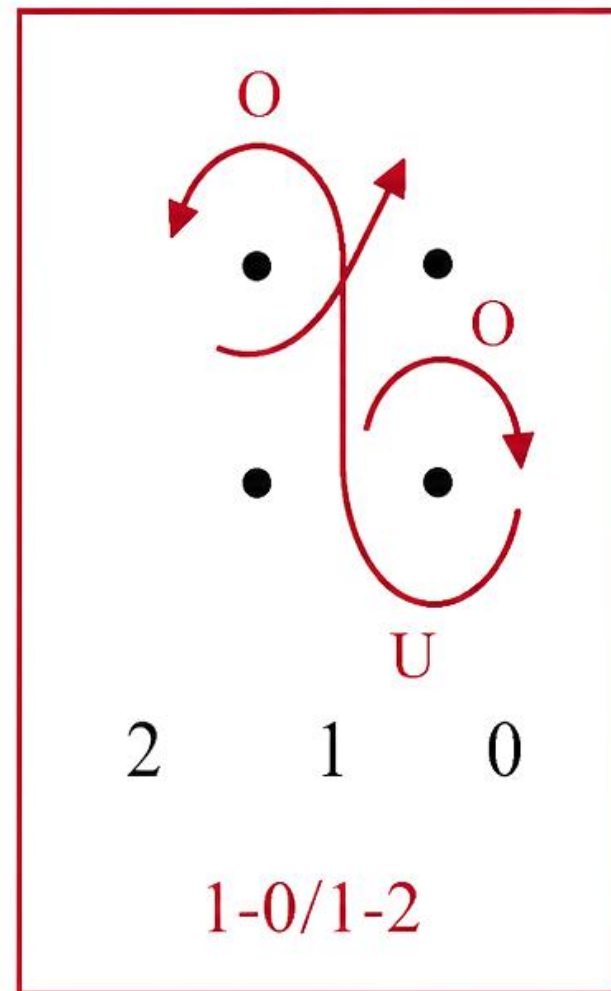
Overlap movements: 1 – 0, 2 – 3

Underlap movements: 0/2, 3/1

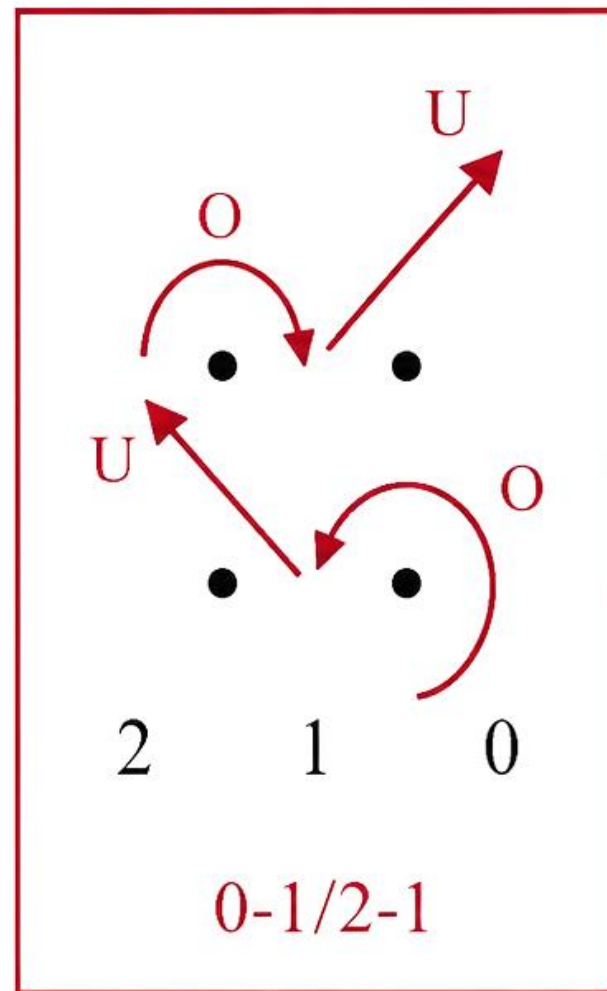
BASIC OVERLAP OR UNDERLAP VARIATIONS

1. Closed Lap: An overlap followed by an underlap in the opposite direction.
2. Open Lap: An overlap followed by an underlap in the same direction.
3. Only Overlap: Only overlap & no underlaps.
4. Only Underlap (Laying-in): Only underlap & no overlaps.
5. Miss Lapping: Neither overlaps nor underlaps.

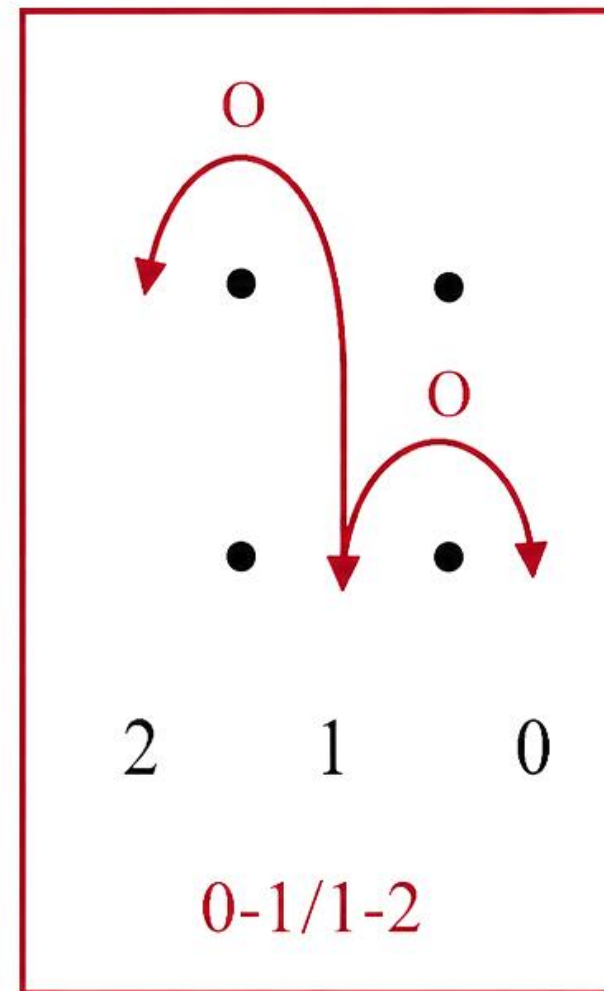
BASIC OVERLAP OR UNDERLAP VARIATIONS



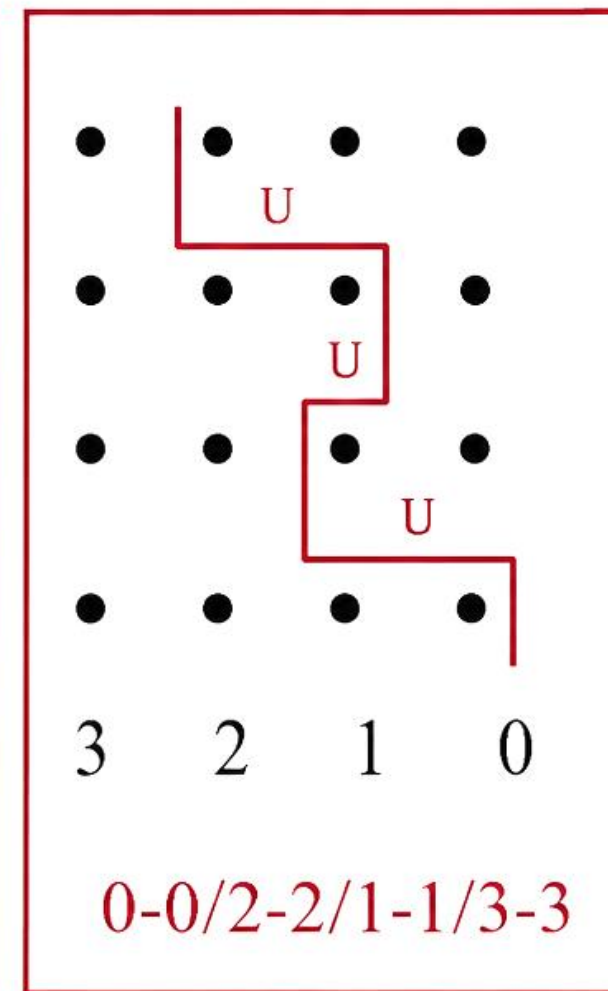
a) Closed lap



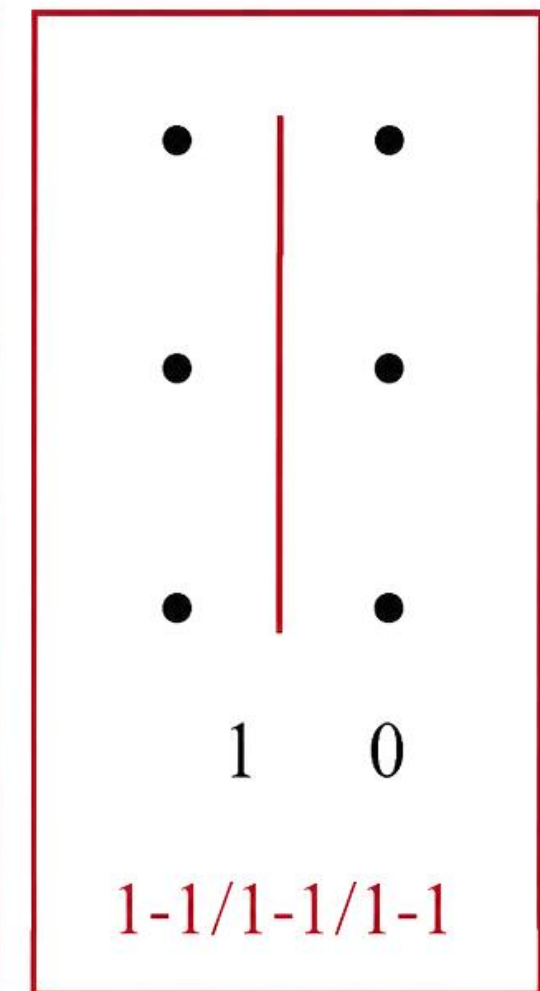
b) Open lap



c) Open lap



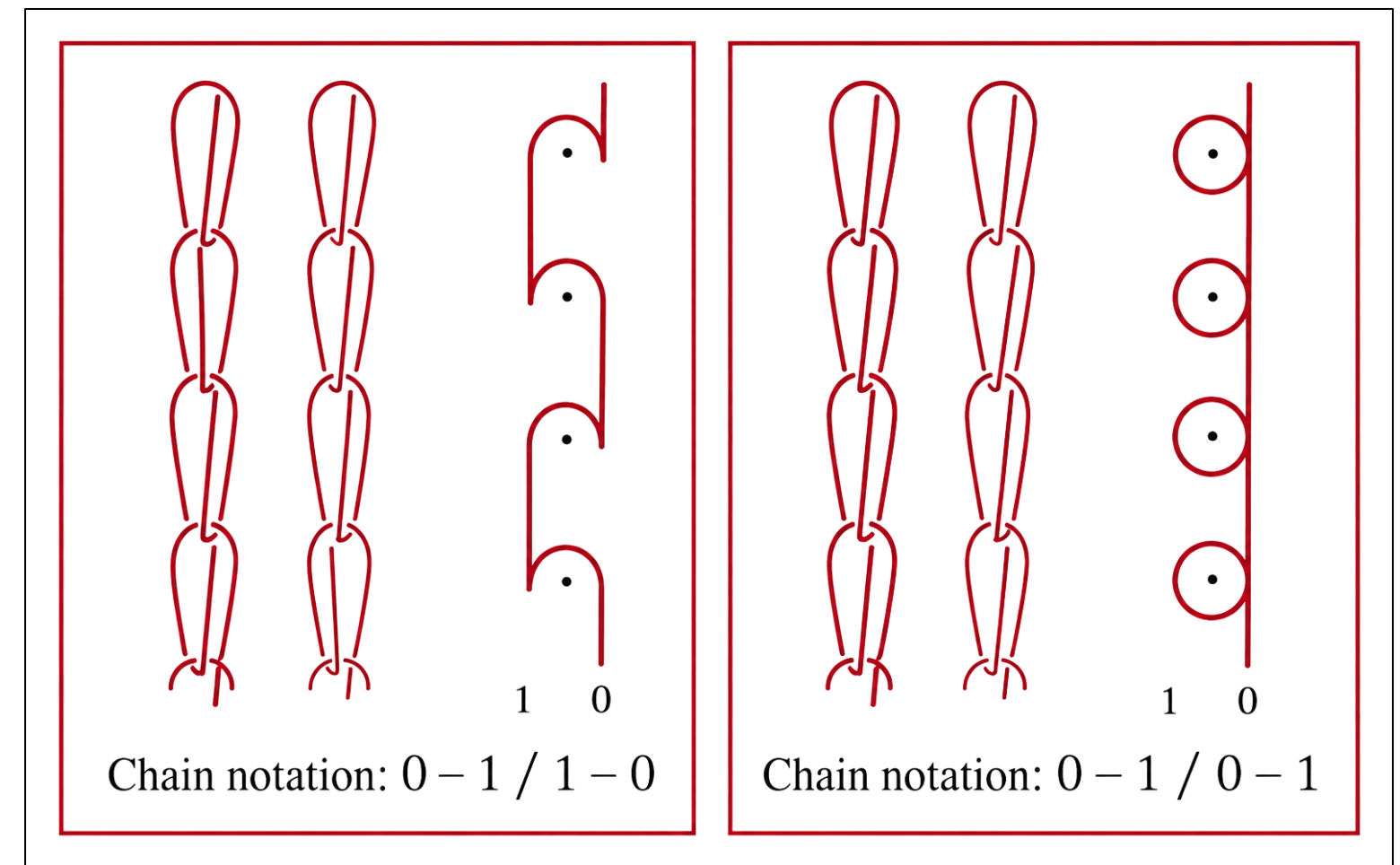
d) Laying-in



e) Miss-lapping

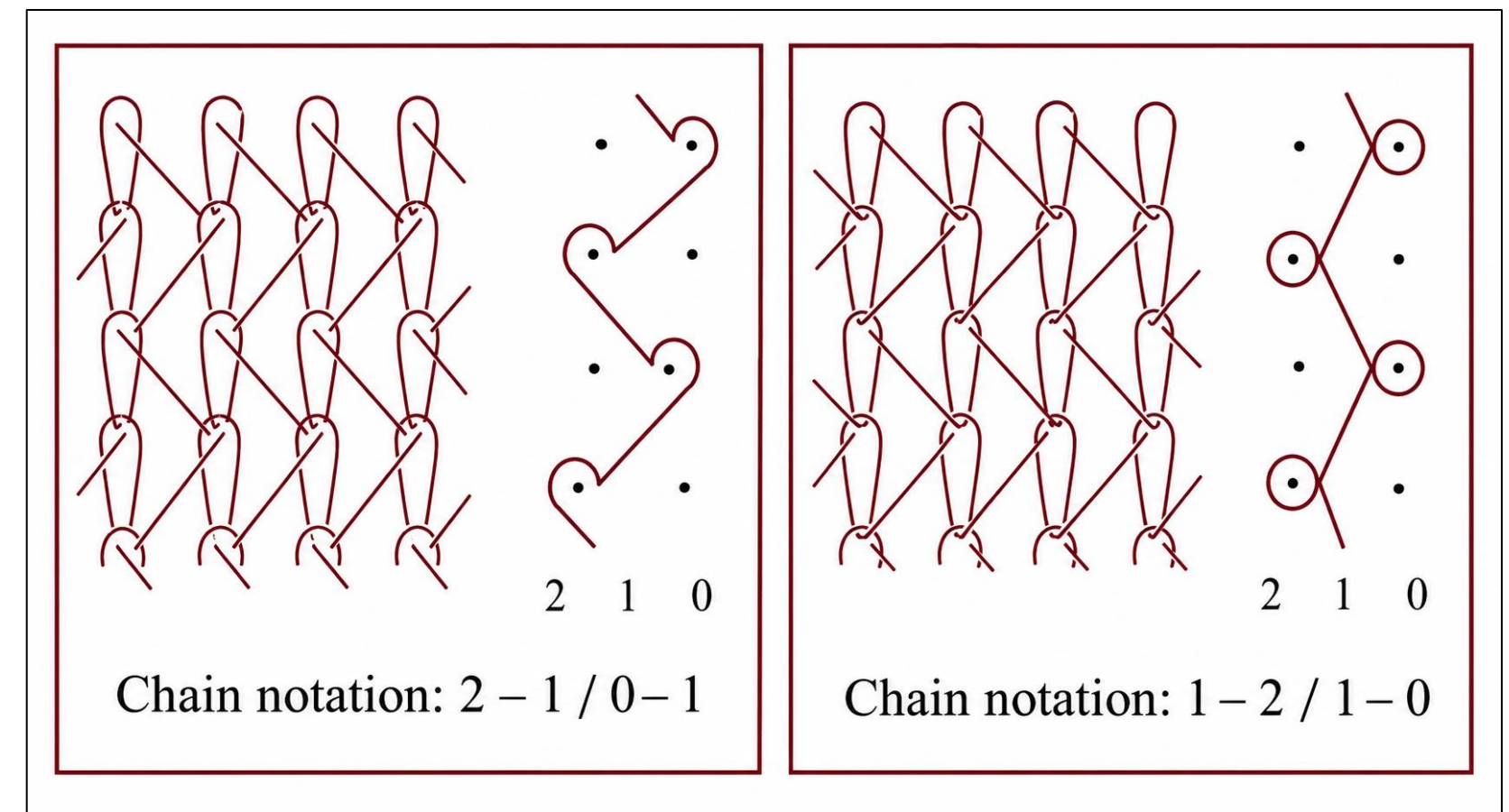
PILLAR OR CHAIN STITCH

- Produced a *vertical chain of consecutively knitted loops on the same needle from the same yarn.*
- The *chain* are separated from each other or connected by other yarns.
- There are *no sideways connection*, so *no fabric* is formed.
- Only wales direction.
- Used to produce *stripe fabric*.



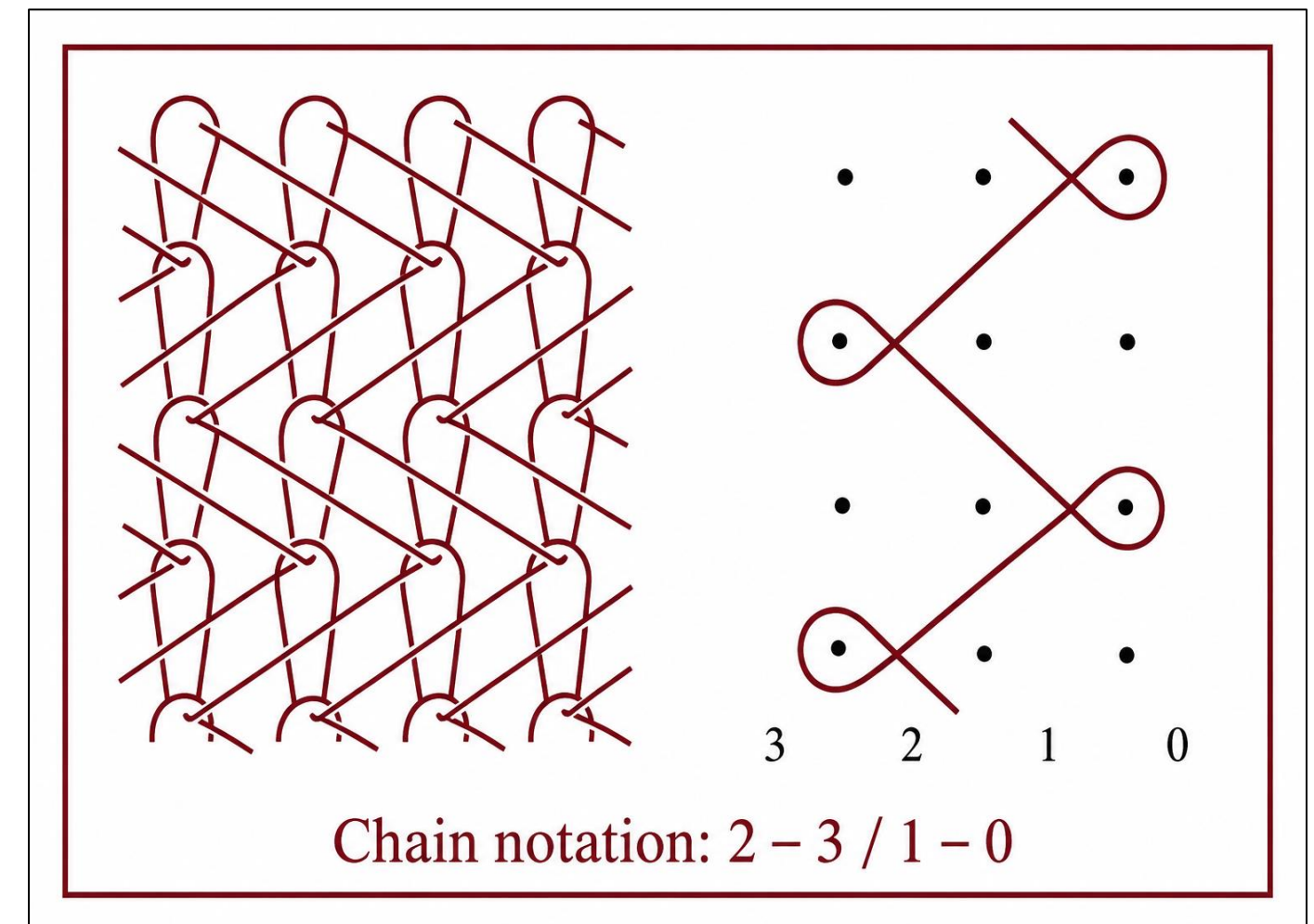
TRICOT STITCH

- Tricot stitch is a stitch formed of one warp; the tricot loops are *disposed in an alternate order in two adjacent wales*.
- The lapping movement is formed by the *guide lapping alternately two adjacent needles*. It is also called a tricot lap and the definition "*1 and 1*" implies an **underlap of one needle space and an overlap of one needle space**.
- *The 1 and 1 movement can be formed open or closed.*
- The closed 1 and 1 structure is more popular.
- The construction is **flexible, light** and very popular in the production of two guide bar fabrics.



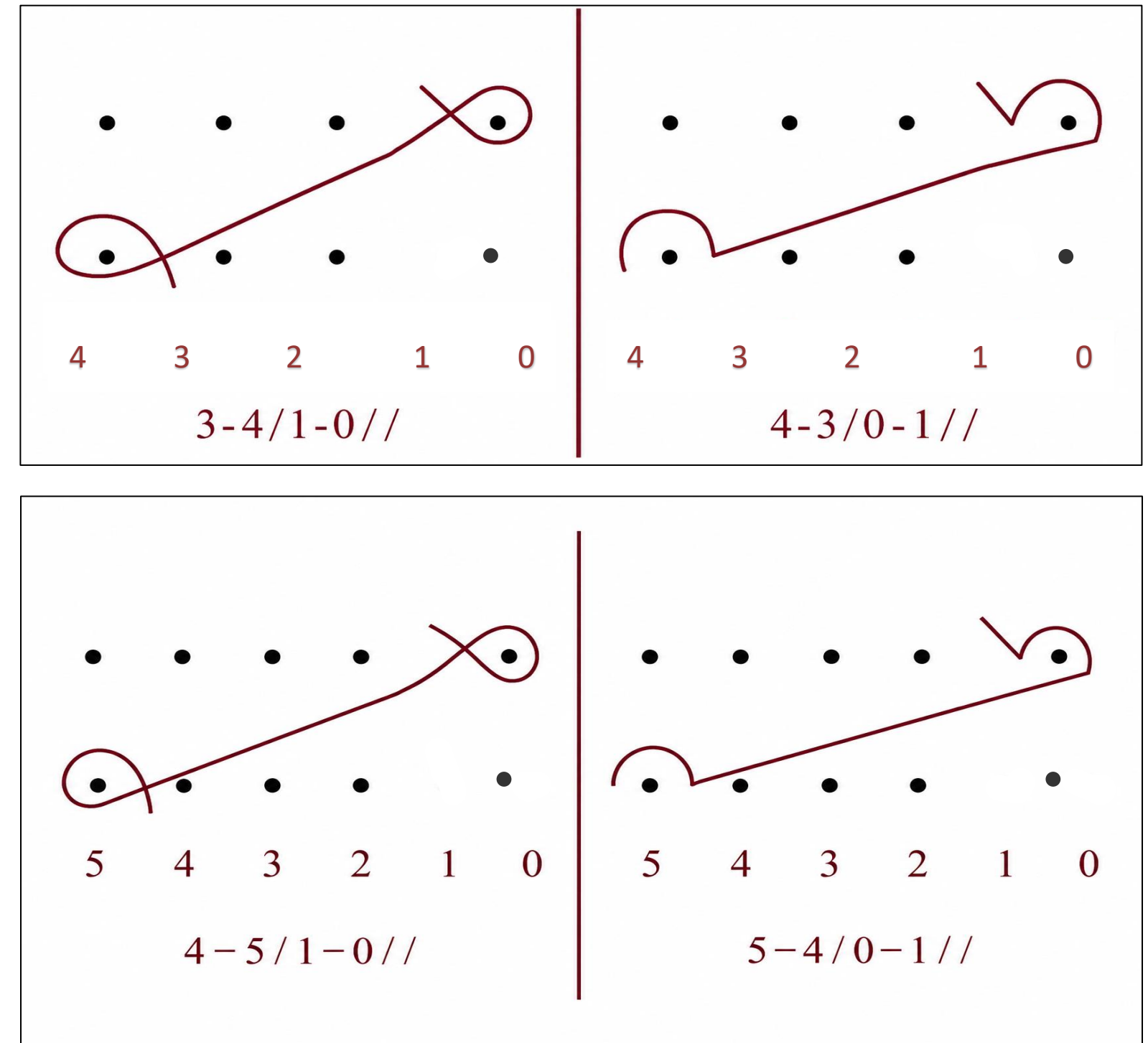
CORD STITCH

- It differs from tricot in the length of links.
- It is knitted from a one warp system and the loops are formed by one yarn in turn in adjacent course, **every two wales**.
- “*2 and 1*” *structure is produced* implies an **underlap** of two needle space and an **overlap** of one needle space.



LONGER RECIPROCATING LAPPING MOVEMENT

- By increasing the underlap shogging movement by one or more needle space (like 3 & 1, 4 & 1) this stitch is formed and it can be produced by open or closed lap.
- These stitches differs from tricot and cord in the length of links.
- *If 3 and 1 lap is constructed then it is called Satin stitch (Above).* More yarn is used with the production of satin stitch. Widthwise stability is increased together with fabric weight and cover factor.
- *If 4 and 1 lap is constructed then it is called Velvet stitch. (Below)*



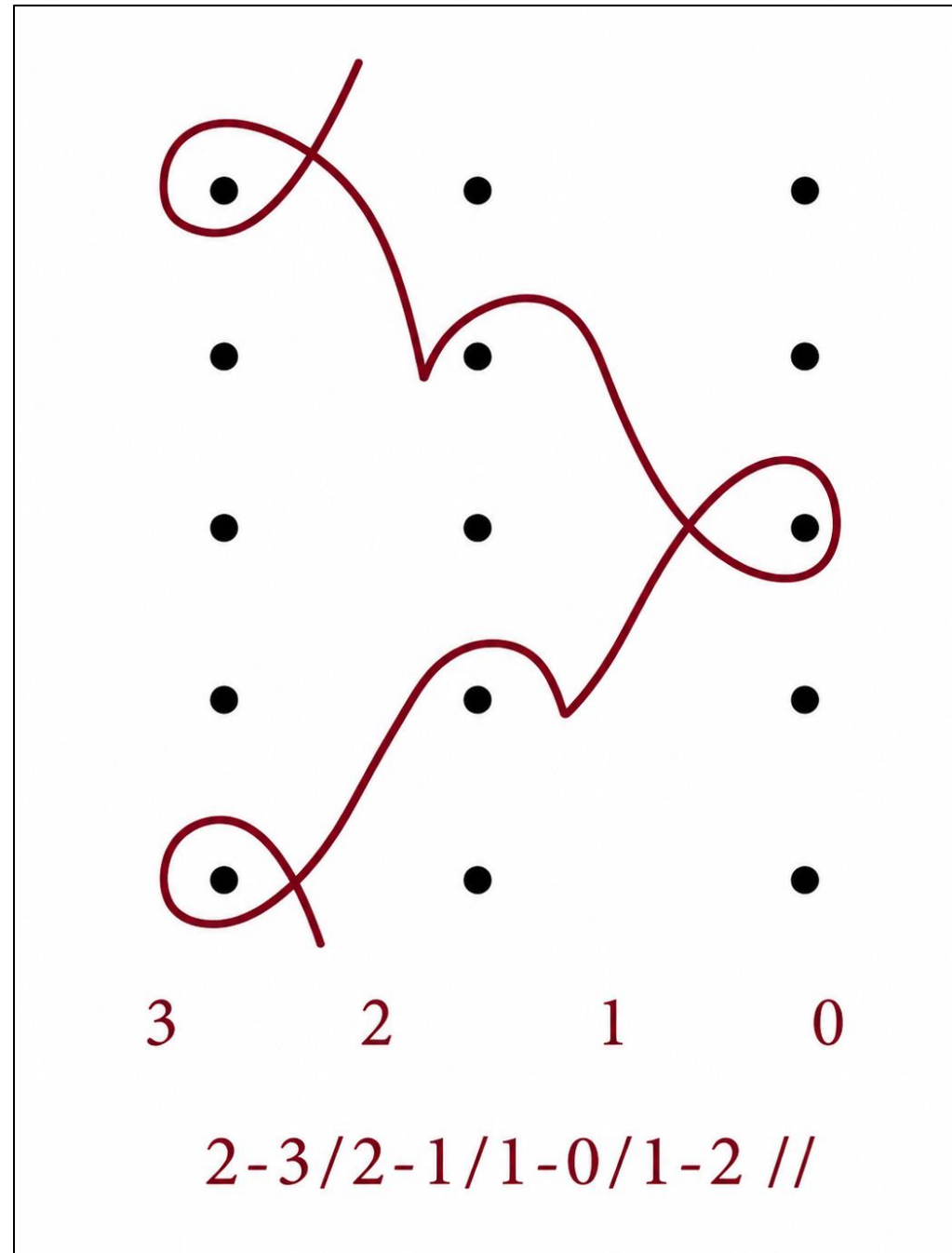
ATLAS STITCH

It is a stitch in which each yarn consecutively forms loops in a **multitude of adjacent wales**. Atlas lapping is often used with the **guide bars threaded with colored yarns**. With two bars moving in opposition, the threading arrangements produce, due to the atlas movement, diagonal or diamond shapes. Atlas stitch is formed by the combination of open and close laps.

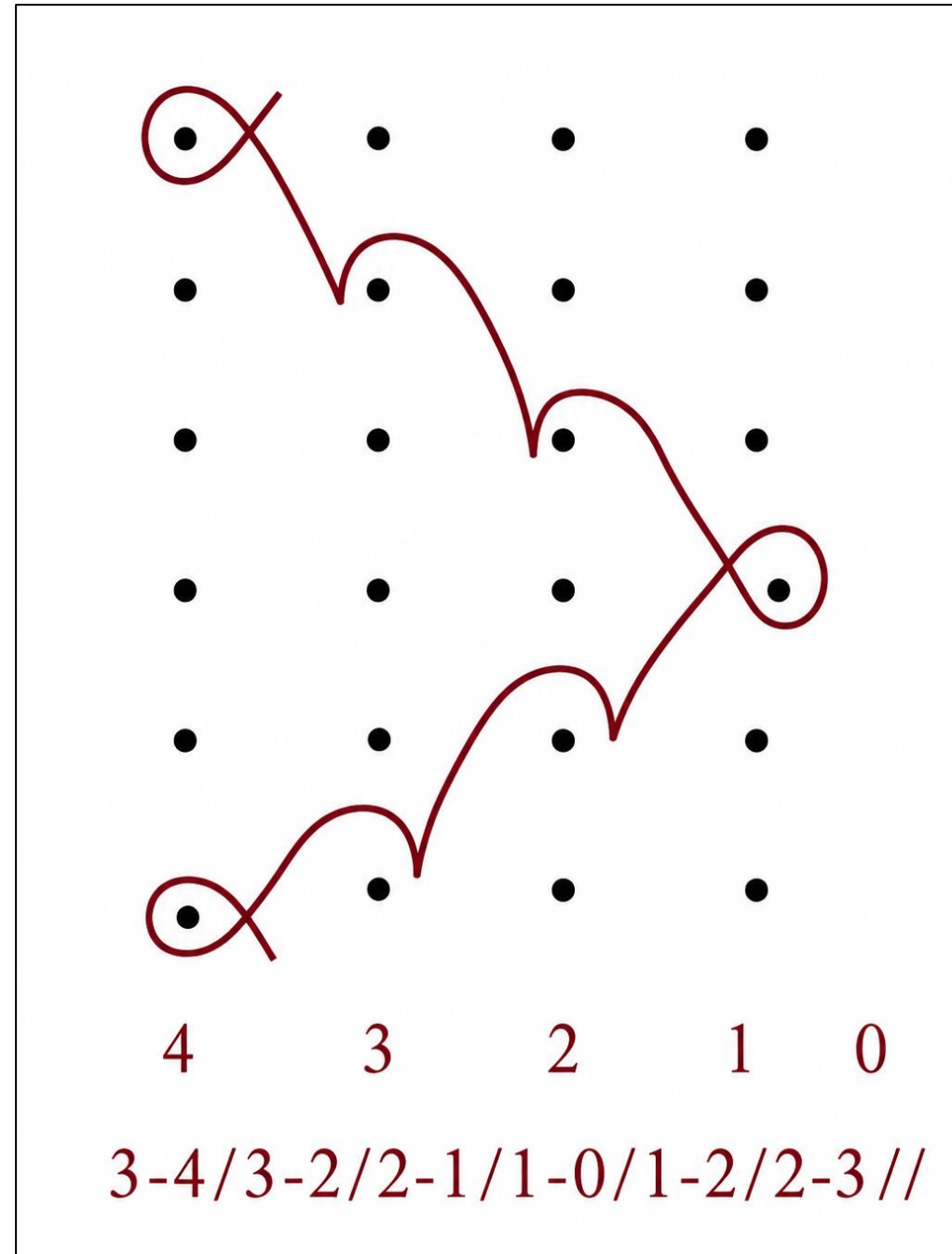
There are various types of atlas:

- (i) Tricot atlas- 2 course tricot atlas, 3 course, 5 course etc.
- (ii) Cord type atlas- It is formed on the **basis of cord stitch**. 5 wales and 4 course in one repeat.
- (iii) Satin type atlas- On the basis of **satin stitch principle**. 7 wales and 4 course in one repeat.
- (iv) Velvet type atlas- On the basis of **velvet stitch principle**. At least 9 wales and 4 course in one repeat.

ATLAS STITCH

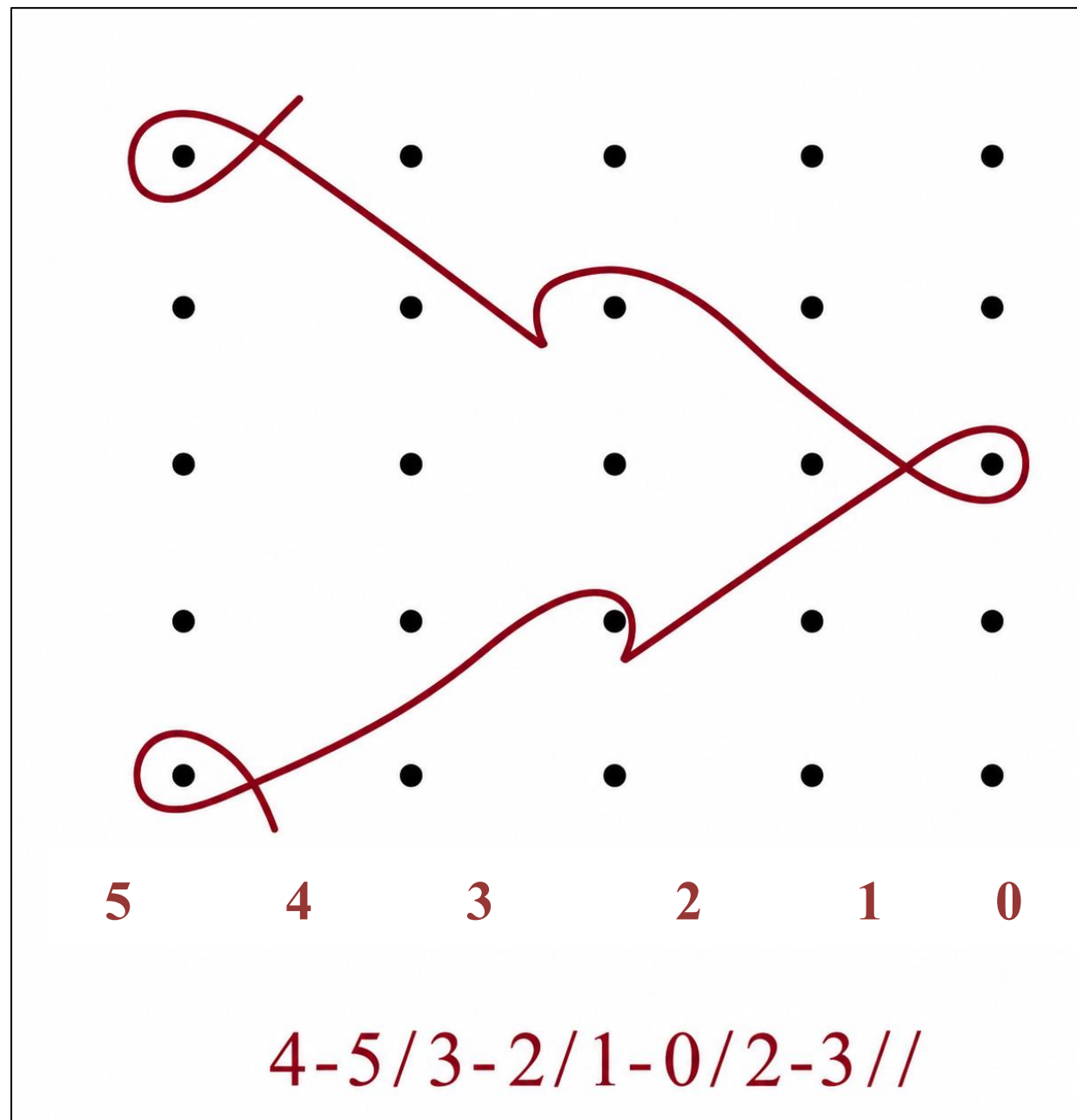


Two Course Tricot Atlas

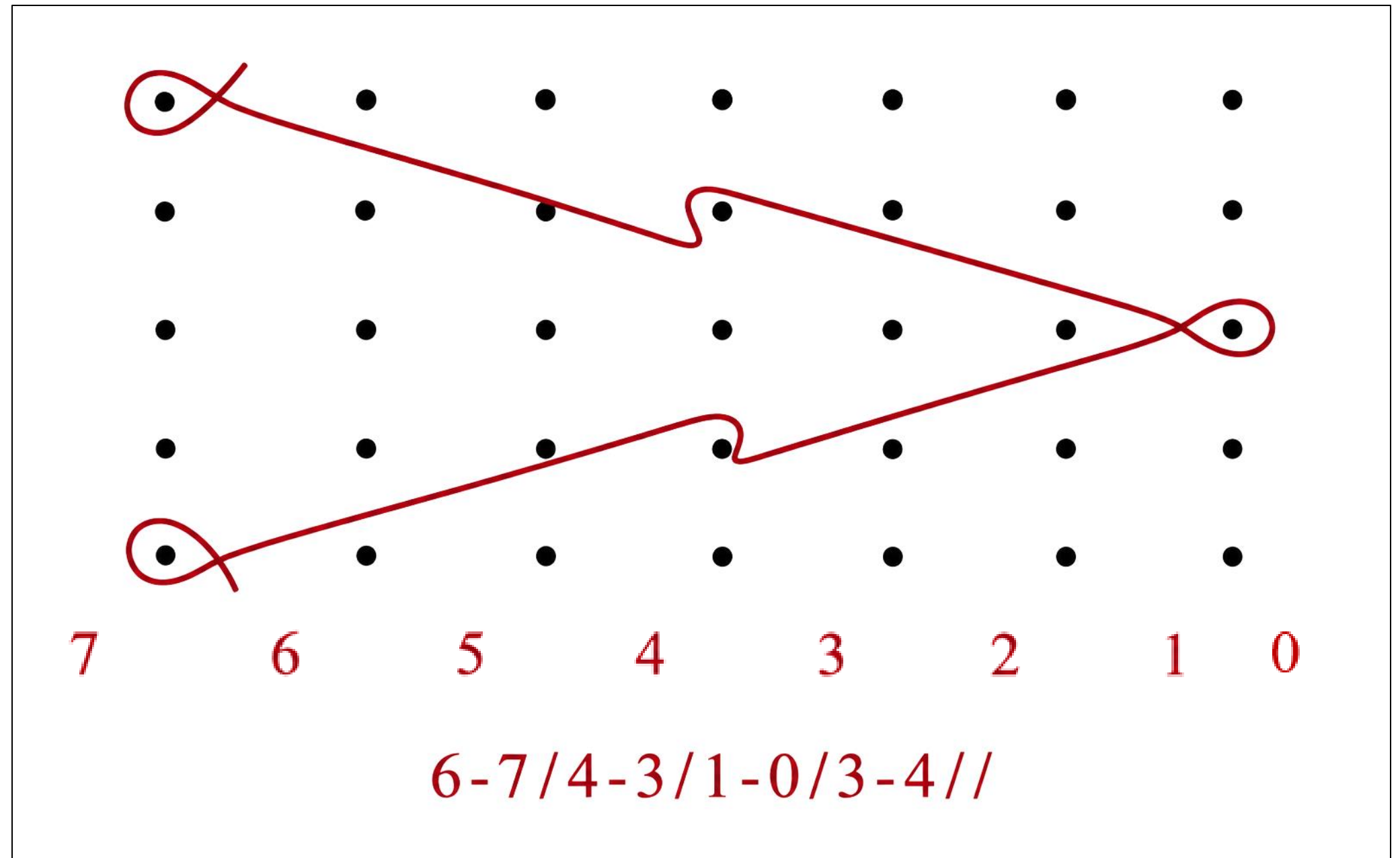


Three Course Tricot Atlas

ATLAS STITCH

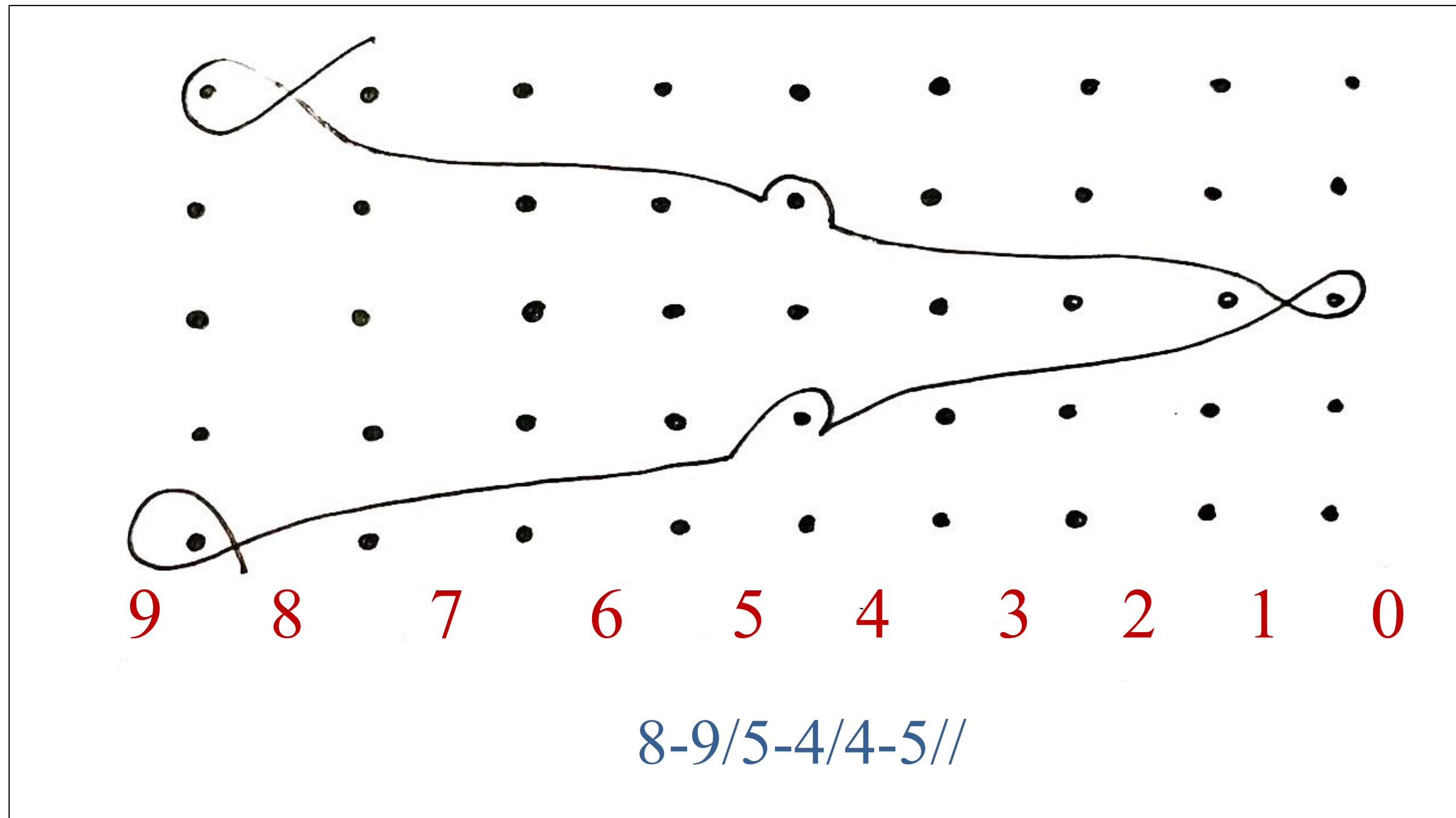


Cord Atlas



Satin Atlas

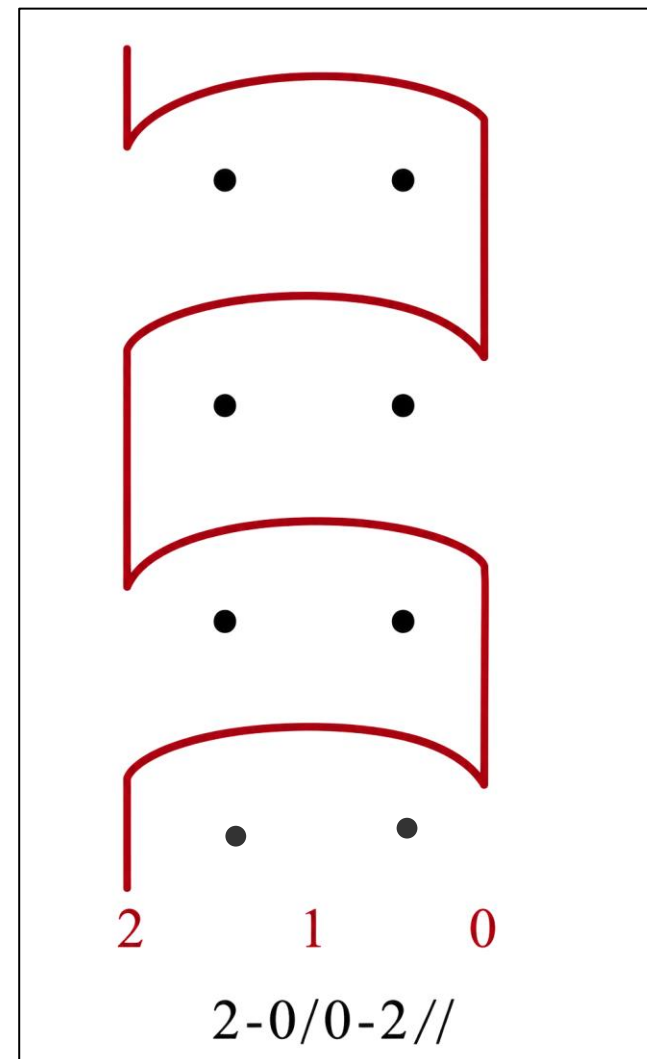
ATLAS STITCH



Velvet Atlas

TWO NEEDLE OVERLAP

- The *guide can be shogged by two needle spaces on the hook side of the needle* during an *overlap*.
- In this way *two needles are wrapped by each yarn* and both will draw the loops simultaneously.
- Used for producing body and stability to a single guide bar fabric.



THANK
YOU !

